ASSESSMENT OF THEMATIC MAPPING TECHNIQUES FOR POPULATION DISTRIBUTION VISUALIZATION: A Comparative study of Dot Density, Filled Isopleth and Unfilled Isopleth

Master Thesis

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Submitted by

Babafemi Sunday Olatunji

Supervisor:

Mag. Dr. Christoph Traun

Department of Geoinformatics Z_GIS

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Author's Declaration

I hereby declare that this thesis is the result of my own work. All sources used in this work have been properly acknowledged and cited in accordance with scientific and academic ethics.

(Salzburg, June 2026)

Abstract

Thematic maps play a critical role in communicating spatial patterns and supporting geographic decision making. Among the various methods used to represent population density, Dot Density maps, Filled Isopleths (Colour-graded), and Unfilled Isopleths (Contour-based) are widely applied, yet their influence on users' perception and interpretation of spatial density patterns remains insufficiently understood. This study investigates how these three cartographic representations affect the perception of population density, with particular emphasis on hotspot identification and spatial gradient interpretation. A between-subjects online experiment was conducted using maps representing both real-world and synthetic population distributions. Participants completed tasks while measure of accuracy, response time, and perceived difficulty were collected. Demographic information and self-reported cartographic experience were also recorded.

The findings demonstrate that thematic map design substantially affects the interpretation of population density patterns. Filled Isopleths appear to provide the most effective support for both hotspot identification and gradient interpretation, while Dot Density maps remain useful for rapid exploratory assessment of spatial clustering. These results contribute to cartographic perception research and provide practical recommendations for selecting thematic map types in population mapping and related spatial analysis applications.

Keywords: Cartographic perception, Thematic Maps, Population Density, Filled isopleth, Unfilled Isopleth, Dot Density, Real-World, Synthetic, Hotspot Identification, Spatial Gradients, Map Usability, Spatial Cognition.

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Abbreviations

DD	Dot Density
FI	Filled Isopleth
UI	Unfilled Isopleth
SYN	Synthetic
RW	Real-World
GD	Gradient
HS	Hotspot

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1. Introduction

Thematic maps are special types of maps that show information about a topic, such as population, income, or temperature. They are important in cartography to visualize spatial patterns and support analysis (Slocum et al., 2023).

The visualization of spatial phenomena plays a critical role in understanding geographic patterns and support decision making processes. Thematic maps are widely used to communicate complex spatial information effectively through appropriate symbolization and map design (Slocum et al., 2023). In the context of natural hazard assessment, thematic mapping techniques can be used to represent population distribution built-up areas, and hazard exposure. Ehrlich et al., (2018) demonstrated the importance of integrating population density and built-up area datasets derived from remote sensing to quantify exposure to natural hazards to multiple spatial scales. Several mapping techniques are commonly used to represent population distribution. Dot Density maps use repeated dots to represent quantities within geographic space, enabling users to visually identify clustering, concentration, and dispersion patterns. Filled Isopleth represent continuous spatial variation through graduated colour surfaces or heatmap-style visualization, allowing smooth transitions between density values. In contrast, Unfilled Isopleth, commonly referred to as isolines or contour maps, use lines connecting points of equal value to represent spatial gradients and continuous phenomena without filled colour regions. Unlike discrete spatial representations, isopleth-based visualization estimates values continuously across geographic space (Traun et al., 2017; He et al., 2011; Slocum et al., 2023). All three techniques are valid; each emphasizes different spatial characteristics and may lead to distinct interpretations by map readers. Previous studies have shown that thematic map design can influence user effectiveness, attractiveness ratings, and overall evaluation of spatial information. Furthermore, different visualization methods may affect how spatial patterns are perceived and interpreted, highlighting the importance of selecting an appropriate mapping technique for a given task. Selecting the most appropriate method therefore plays a crucial role in thematic design (Medynska-Gulij et al., 2020; Besançon et al., 2020; Forrest & Medynska-Gulij, 2021).

The effectiveness of these thematic mapping techniques is closely related to cartographic communication and spatial cognition, as different visualization methods require user to interpret spatial information through varying levels of abstraction and visual complexity (MacEachren 2004). However, different thematic mapping techniques emphasize different characteristics of spatial data. Because these visualization methods communicate information in different ways, they may influence how users perceive, interpret, and evaluate spatial patterns.

1.1 Map Interpretation and User-Centred Design

Research has shown that the way information is represented on a map influences not only what users see but also how they understand and interpret spatial relationships (MachEachren, 2004).

Thematic maps are used to show spatial information in an abstract, classified, and symbolized way and are able to represent complex geographic phenomena in such a way that facilitates pattern recognition and interpretation (Ding & Meng, 2014). In modern cartography, a user-centred approach is used, in which perceptual and cognitive issues are considered as important factors in map usage and interpretation (Roth et al., 2017).

Users' interpretation of maps is not always consistent, as shown in recent studies. The performance of map reading is affected by visual features of a map as well as person and task features of the user. There is evidence that the design of maps influences how accurately and efficiently users can extract and interpret the spatial information from a map (Pickle, 2004). In addition, there is no one map design that is best for everyone or for any particular analytical problem, which emphasizes the need to consider the user-centred perspective of evaluating cartographic visualizations (Pickle, 2004). Likewise, Medynska-Guly et al. (2023) demonstrated how user testing can provide valuable insights into the practical effectiveness of thematic maps and the disconnect between users' preferences and their ability to perform better with the maps.

The results of Beitlova et al. (2020) also show that map design has a significant impact on the understanding of the map and task achievement. They found that the users have different levels of difficulty when they use different thematic maps; the difficulty depends on how they interpret the symbols, how they use the legend, and how complex they are. The authors pointed out that a good map design should assist the user to recognize patterns, find information, and do analysis quickly and accurately. Therefore, the assessment of various thematic mapping techniques by means of accuracy, response time, and user perception are necessary to understand the effectiveness of these techniques in the communication of spatial information.

All of these studies indicate that careful attention must be given to control the amount of visual information and complexity of a thematic map. A good map will be informative, without being too complicated and will be easy to read. This assists people to accomplish their jobs.

Identifying places, comparing or contrasting, and figuring out patterns without getting lost. Researchers can determine which maps provide the optimal user satisfaction and level of accuracy through testing and comparing with various types of maps.

1.2 Research Gap

Several works have explored the concept of thematic mapping and the way that it can be displayed on maps, while a few have looked at comparative user-based evaluations between Dot Density, Filled Isopleth (colour graded) and Unfilled Isopleth (contour lines) maps in the same experimental setting. Recent studies have shown that the use of different thematic map types can lead to marked differences in map-reading and in the interpretation of the map information. Sukraini et al. (2022), for example, examined choropleth maps and graduated symbol maps for communicating the distribution of COVID-19 cases in Bali, Indonesia. In total, 120 people completed eleven tasks on the map, such as identifying, comparing, interpreting, categorising and locating spatial information. Three factors were measured in the study: accuracy response time, and the level of difficulty. The results indicated that participants who were presented choropleth maps had the highest accuracy (73%) and those who were presented graduated symbol maps had the lowest accuracy (63%). Further, choropleth maps also led to faster task completion times, with the average response time being 28.15 seconds while the graduated symbol maps were 35.78 seconds. Participants also felt choropleth maps were more readable and understandable. The results of this study led the authors to believe choropleths maps were better at communicating spatial patterns and supporting map-reading tasks than graduated symbol maps. The same results were obtained by Słomska-Przech and Gołębiewska (2021) in a large-scale user study of 366 participants comparing choropleth, graduated symbol and isoline maps. A series of 11 map-reading tasks to test various aspects of the processes of spatial information processing, such as identification, comparison, interpretation, and pattern recognition. Study variables included accuracy of answers, completion time, and perceived difficulty of the task. The results showed the highest overall accuracy choropleths maps had the shortest response times and were easiest to use of all types of maps. But isoline maps tended to be more difficult, more time consuming to answer, and less accurate. The authors found that the type of thematic map can have a great impact on objective performance and subjective user experience, and that some maps are more appropriate for specific analytical tasks than others.

Overall, these studies suggest that the cartographic representation affects the perception and understanding of spatial information by users. The previous studies, however, have been mostly concerned with comparisons between choropleth, graduated symbol and isoline. Thus, the relative efficiency of mapping spatial distribution patterns by Dot Density, Filled Isopleths and Unfilled Isopleths maps under both controlled synthetic and realistic conditions is not yet known. This study is thus an extension of previous cartographic perception studies, as it compares the different map types in terms of objective performance measures (accuracy and response time) and subjective measures of perceived difficulty.

1.3 Aim of the study

This research aims to fill that gap by testing these three map types using both real-world population data and simulated datasets. It examines how each technique affect user' ability to recognize population patterns such as hotspot and gradient effectively.

1.4 Research Objectives

1.4.1 Main Objective

To assess the impact of different thematic mapping techniques on the interpretation of spatial population distribution patterns.

1.4.2 Specific Objectives

1. To compare the effectiveness of Dot Density maps, Filled Isopleths, and Unfilled Isopleths in hotspot identification tasks.
2. To evaluate how different thematic mapping techniques support spatial gradient interpretation along continuous population profiles.
3. To compare interpretation accuracy and response time across thematic map types.
4. To assess perceived cognitive difficulty associated with hotspot and gradient tasks.
5. To investigate the relationship between subjective task difficulty and objective interpretation performance.
6. To evaluate thematic map usability using both synthetic and real-world population datasets.
7. To provide recommendations for selecting appropriate thematic mapping techniques for population visualization and spatial decision-making tasks.

1.5 Document organization

The design of the study is described in the subsequent sections, comprising sources of data, simulation process, methods to develop and test thematic maps. Then, I show the outcomes of the comparative analysis of the map techniques, in terms of user performance. Finally, I sent out the major results of the study and make recommendations.

2. Methods

In this study, the effectiveness of the three thematic mapping techniques (Dot Density (DD), Filled Isopleth (FI) and Unfilled Isopleth (UI)) in assisting the interpretation of spatial distribution patterns for the population is assessed using a controlled web-based user experiment. The research is centered on two main tasks of spatial interpretation, namely identification of hotspots and spatial gradient interpretation. The experiment compares the effectiveness of the maps by measuring interpretation accuracy, response time, and task difficulty. The metrics applied are similar to the widely used usability metrics for cartographic usability studies and thematic map evaluation studies, such as effectiveness (correctness), efficiency (time) and subjective task difficulty (Çöltekin et al., 2009; Słomska-Przech and Gołębiewska, 2021).

2.1 Experimental Design

A between-subjects experimental design was adopted to investigate performance difference among the three thematic mapping techniques. In alignment with user-centred design frameworks in cartography usability (Roth, 2013), participants were exposed to only one cartographic representation throughout the survey. The approach minimizes potential learning effects, prevents for carry-over effects, visual familiarity effects, and participant fatigue that may occur when users interact with multiple visualization conditions during same experiment (Lazar et al., 2017). This structure ensures that performance variances can be directly attributed to the cartographic design itself rather than past task exposure (Roth, 2013).

The between-subjects experimental design was deemed suitable as the study aims to compare cognitive interpretation differences in two different cartographic representations in controlled environment.

The independent variable (IV) in this study is the thematic visualization techniques with three experimental conditions:

Dot Density maps (DD)

Filled Isopleth/Colour graded maps (FI)

Unfilled Isopleth/Contour lines maps (UI)

For random assignment of the participants to the three experimental conditions, a randomization procedure was used within LimeSurvey.

In this study, dependent variables (DVs) measured are:

- Interpretation Accuracy: Logically, the metric is binary (0=incorrect, 1=correct) reflecting the participant's capacity to correctly identify hotspot regions, and accurately interpret gradient patterns within thematic maps.
- Response Time: The precise amount of time (In seconds) for which the response was made after the presentation of the first map stimulus. This temporal metric can be an objective indicator of cognitive efficiency and cognitive processing speed in map reading workflow.
- Perceived Difficulty: Assessed by a post task 5-point Likert scale for each task. A subjective scale that reflects user perceived cognitive workload, reflecting how easy or difficult the participants perceived the processes of interpretation of the hotspot and gradient.

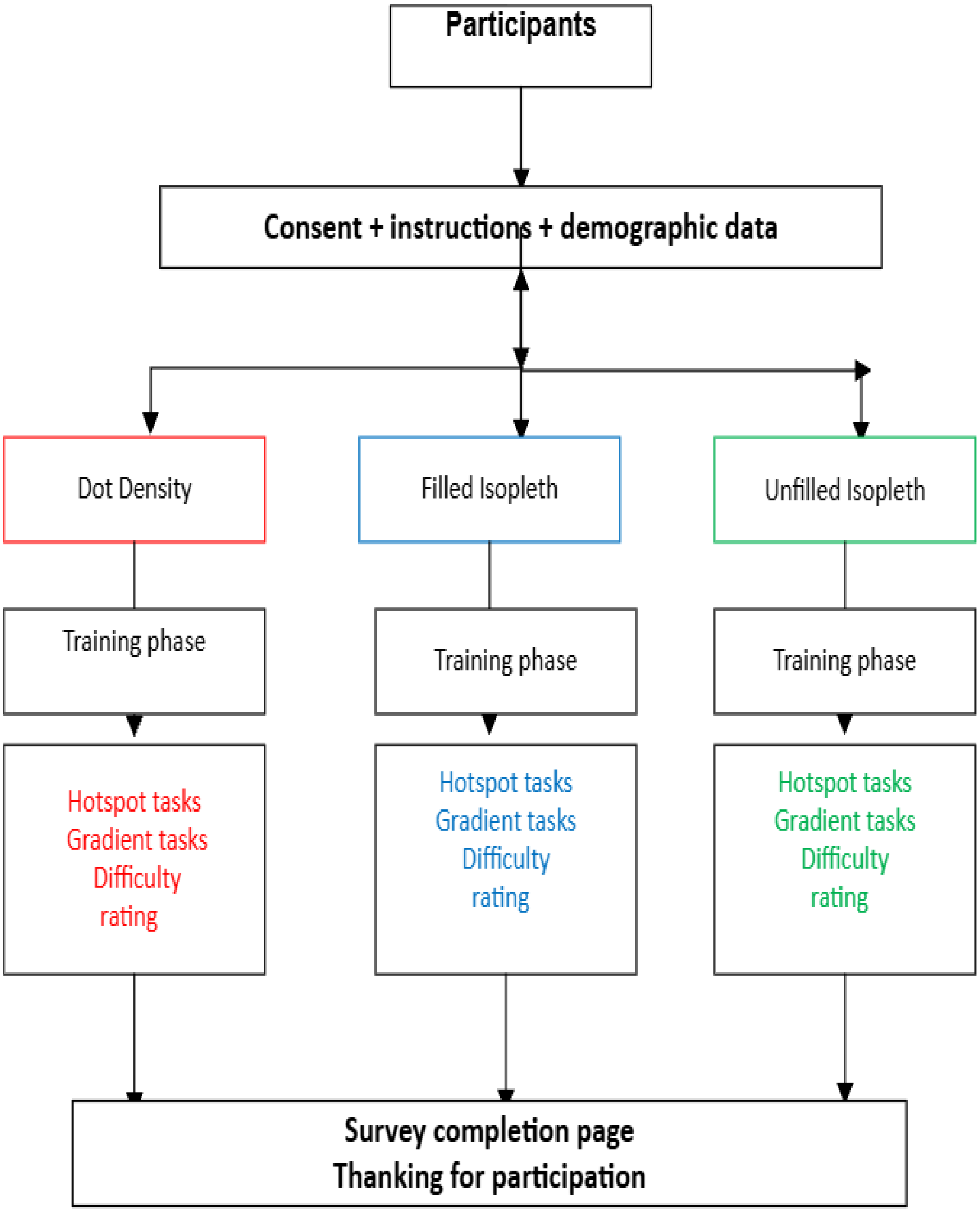


Figure 1: Graphical overview of the experimental workflow and random assignment procedure used on Limesurvey.

2.2 Stimuli Generation

The maps were created using actual real-world population data alongside simulated population data. All the thematic map variants were created from the same data sets. The layouts, boundaries, scales and visual frameworks were exactly the same for all three mapping styles. The strict standardization helps to externalize the mapping techniques, removing design biases in the outside to measure the various styles of mapping techniques.

2.2.1 Real-World population Datasets

Two study areas were chosen in the real world that showed different spatial patterns of populations. The Ruhr Region in Germany was selected as the area of a density gradient because it is highly urbanized and polycentric – meaning that many cities begin to merge into a continuous urban landscape, with a gradual transition to the adjacent rural landscape. The selected areas were Essen, Dortmund, Bochum, Duisburg, Gelsenkirchen, Recklinghausen and the neighbouring districts. Population data were obtained from the worldPop database, while administrative boundaries were derived from GADM database.

Highly clustered population hotspot patterns, where there are dense urban areas with pockets of low-density areas, were represented in California, United States. This dataset was largely utilized for the purpose of identifying hotspots and was also provided from the same database.

Thematic map interpretation under realistic geographic conditions was evaluated using real world population data sets.

2.2.2 Synthetic Population Datasets (Controlled Spatial Patterns)

The synthetic population datasets were created using procedural simulation techniques in Python. Some parts of the script and workflow were developed with the help of AI coding tools. These datasets were produced to produce controlled distribution patterns to isolate certain cartographic interpretation problems. Precise control of density, spatial arrangement, gradient transition and complexity of distribution was achieved by the synthetic approach. The four main synthetic spatial architectures obtained were: the clustered and the density gradient distributions. Constructed datasets were generated in the artificial polygonal basemap adapted from the previous research of cartographic visualization (Traun., 2021) to provide a neutral spatial framework for the generation of synthetic population patterns.

Each point in a population distribution was plotted separately. This systematic approach ensured that the same underlying spatial phenomena would be converted in the same manner using the three thematic mapping techniques. The use of synthetic datasets provides better experimental control and enables direct comparisons of the performance of the thematic maps under predetermined spatial conditions.

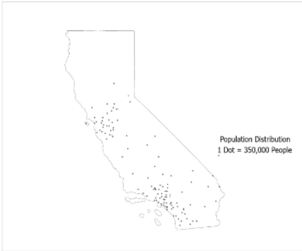
2.2.3 Cartographic Implementation

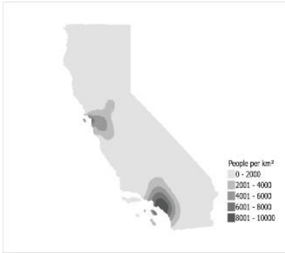
All thematic maps were created from the same basic data and processed with the same cartographic parameters, so that any differences in interpretation is more likely to be due to the nature of the thematic representation than to differences in basic data. Repeated point symbols distributed by location in a spatial pattern defined by population density were used to create dot density maps. All values of the dots and symbol sizes are standardized to ensure that they are of the same visual dimension as the continuous surfaces. Using Kernel Density Estimation (KDE), Point Based population datasets were converted to continuous surfaces to create filled and Unfilled Isopleths. Filled isopleth maps were created using a colour gradient for the different population densities across the geographical space. To avoid visual domination and in order to limit possible perceptual bias compared to the other thematic mapping techniques employed in the study, a neutral Gray-light-to-dark colour scheme was intentionally used. Contour lines that represent equal-density intervals between the surfaces were used to create Unfilled Isopleths. The upper-class values from the classification scheme used for Filled Isopleths were used as the Unfilled Isopleths' contour intervals to ensure visual consistency and comparability between the two types of maps. This method alleviated over-density of contour lines, made the maps easier to read, and allowed for a fair visual comparison among the various thematic mapping methods employed in the study. All thematic maps were standardised to ensure consistency and fairness over the experiment, having the same map extent, the same layout design, the same display conditions, the same legend content and positioning, and the same visual hierarchy.

Total of 18 maps were created and served as the stimuli to be used when solving different map use tasks (Figure 2).

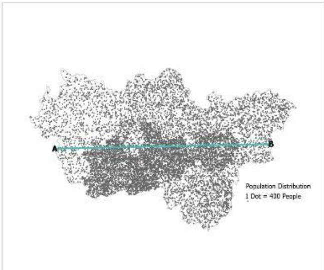
DD **FI** **UI**

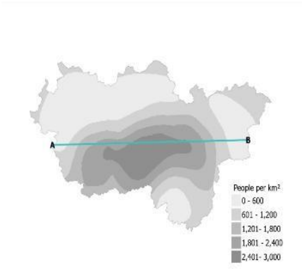
Real-world Hotspot

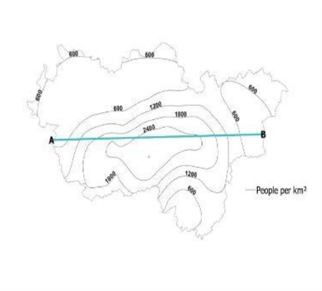




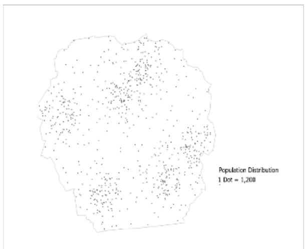
Real-world Gradient

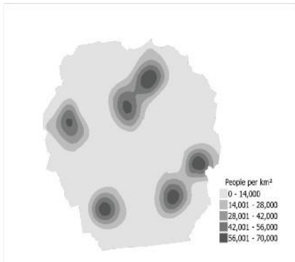




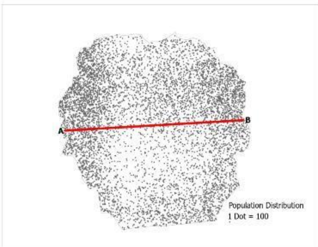


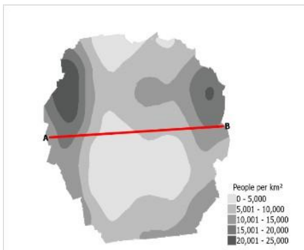
Syntheti
c (A)
Hotspot

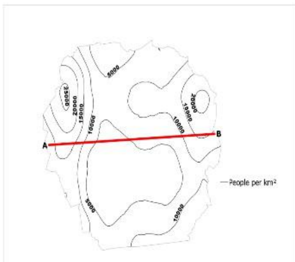




Syntheti
c (A)
Gradient







Syntheti
c (B)
Hotspot

Syntheti
c (B)
Gradient

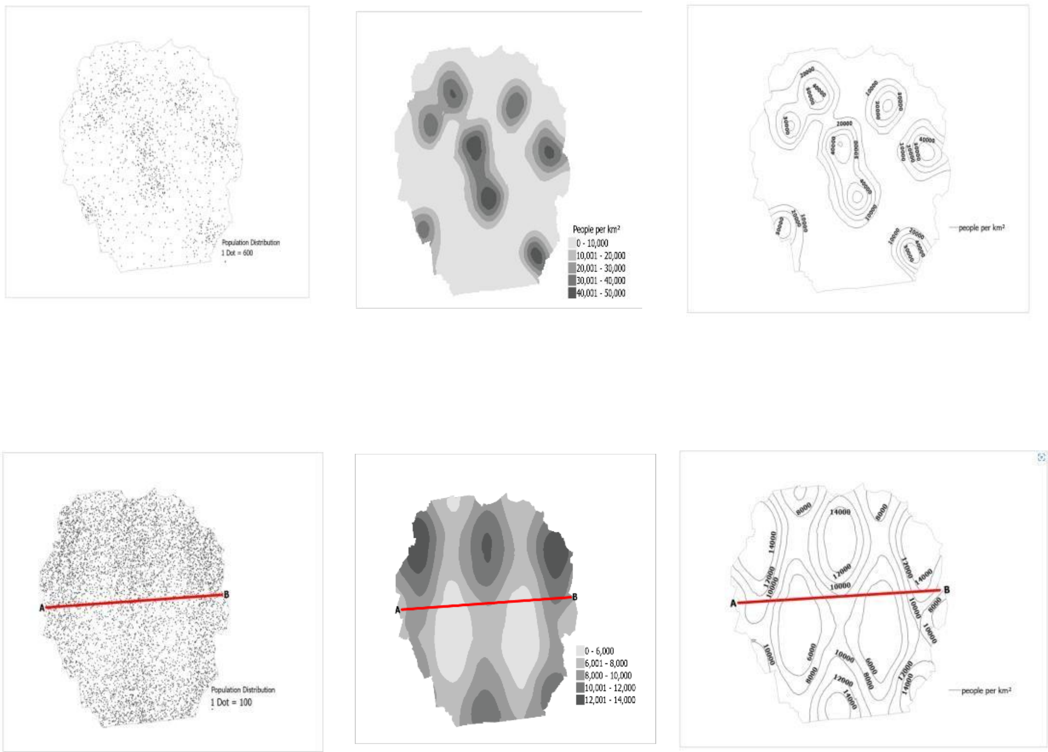


Figure 2: Map types assessed: dot density map (DD), filled isopleth map (FI), Unfilled isopleth map (UI)

2.3 Task Taxonomy

The visualization group did not affect the participants' performance on the spatial interpretation tasks. The experimental tasks aimed at assessing two key elements in the interpretation of a thematic map: hotspot recognition and spatial gradient interpretation.

2.3.1 Hotspot Identification Task

The term hotspot is used to refer to high population distribution areas that visually emerge from the surrounding, although in a spatial analysis sense it may be more appropriate to use the term clusters. Hotspot was retained in this study because it was consistently used throughout the survey design, task description, stimulus labelling and subsequent analysis.

The hotspot identification task was designed to assess participants' capacity to identify the number of hotspots in thematic maps, Participants were asked to identify how many different hotspot areas they could see on each map stimulus. This is an assessment of cluster recognition, density concentration perception and visual pattern interpretation in the three.

thematic mapping techniques. For each of the thematic map type, instructions and specific practice examples were given.

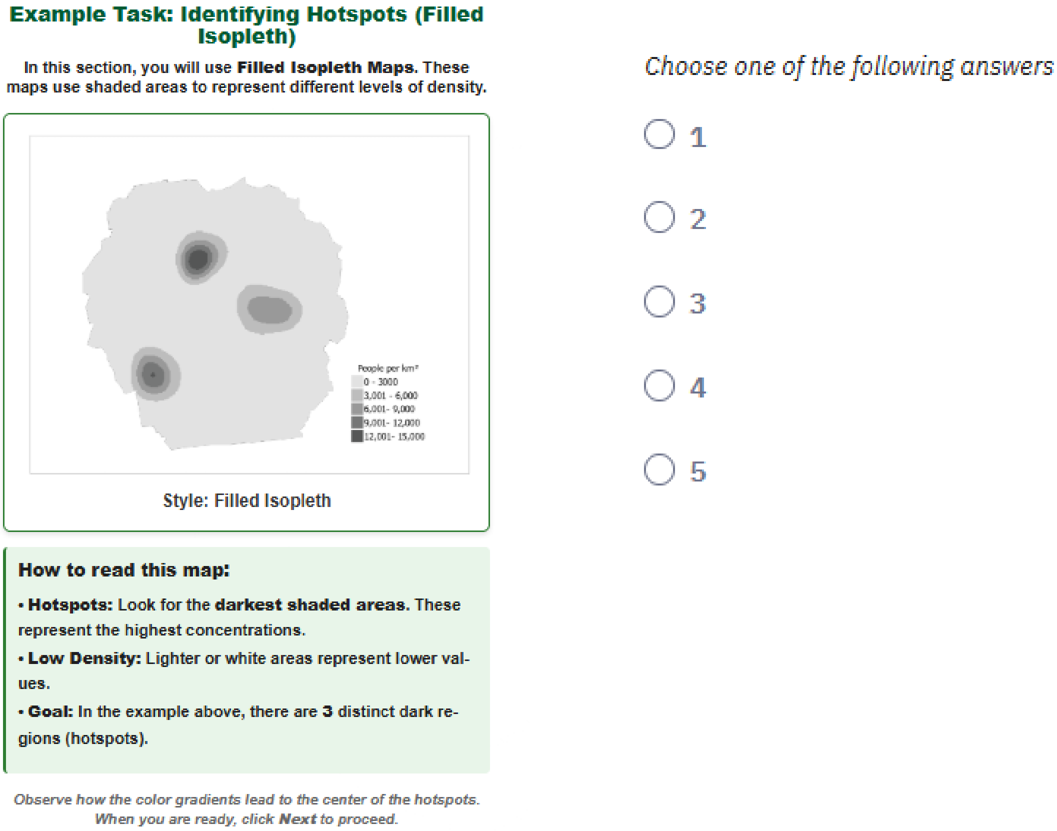


Figure 3: One of the Practice examples for hotspot identification

2.3.2 Spatial Gradient Interpretation Task

Spatial gradient interpretation task was used to assess the ability to interpret continuous spatial population density changes across the geographic space. The participants were asked to look at changes in density along a given line from A to B and pick a graph which matched the observed variation in density. This task measures spatial reasoning, interpretation of gradients and continuous surface perception.

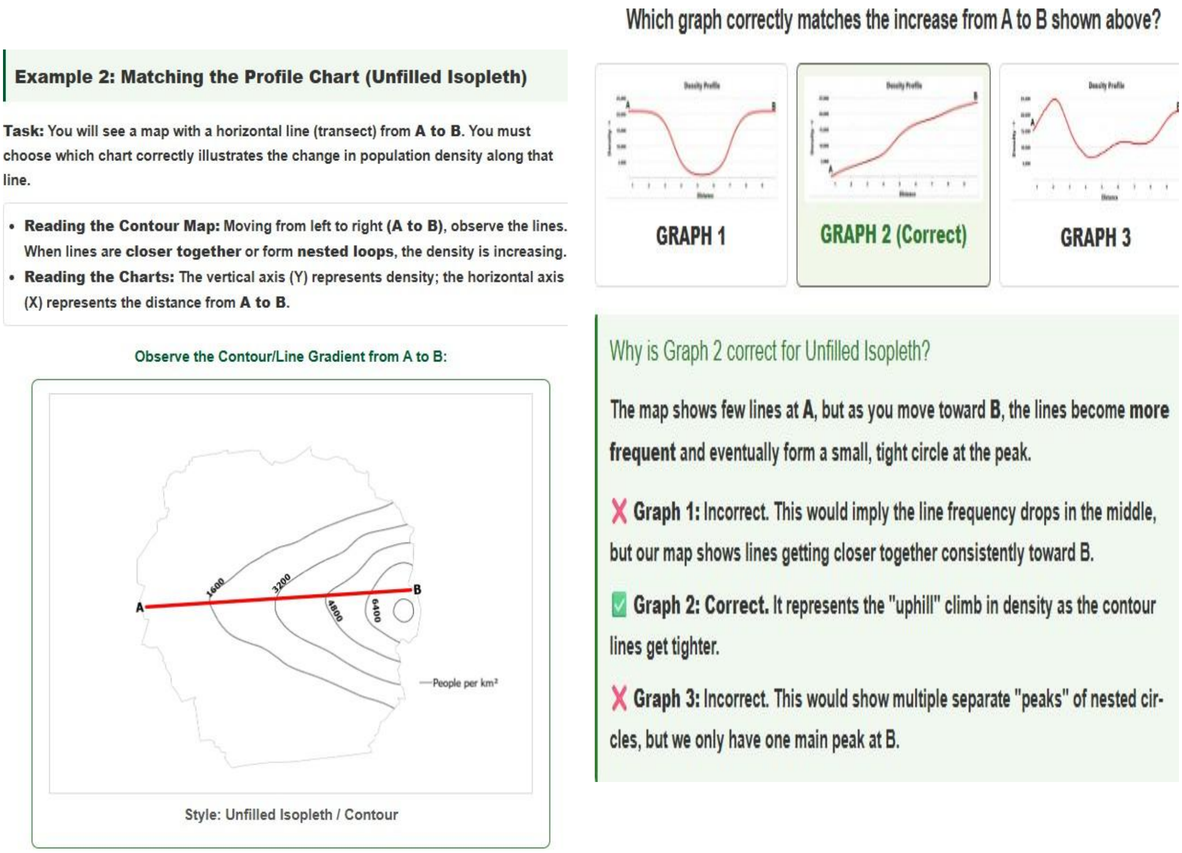
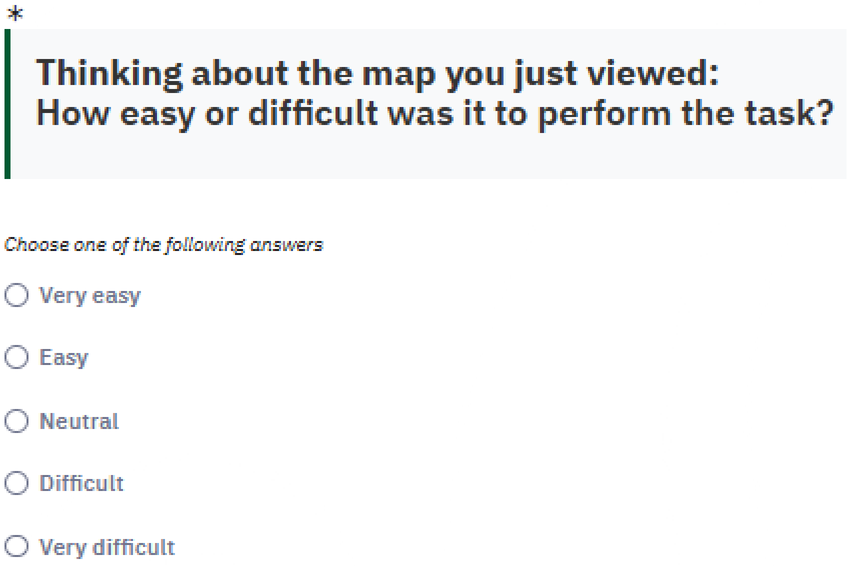


Figure 4: One of the practice examples for gradient interpretation

2.3.3 Subjective Difficulty Assessment

After every experimental task, the participants were asked to rate the difficulty or ease of the task. This subjective difficult assessment was introduced to explore the cognitive load felt by the participants and to determine the ease of use of the thematic maps when using different visualization techniques.

These were subsequently compared with objective data like interpretation accuracy and response time.


*

Thinking about the map you just viewed:
How easy or difficult was it to perform the task?

Choose one of the following answers

- ◯ Very easy
- ◯ Easy
- ◯ Neutral
- ◯ Difficult
- ◯ Very difficult

Figure 5: Difficulty level task

2.4 Participants and Data

The main study was conducted through Limesurvey distributed via academic networks and social media platforms. Invitations were shared with university students, geoinformatics-related communities, and personal contacts with the request for further dissemination. To improve participant diversity and achieve more balanced experimental groups, additional responses were collected through Clickworker platform. Online crowdsourcing platforms have been widely used in cognitive behavioural, and usability research due to their ability to provide diverse participants samples and reliable experimental data (Crump et al., 2013; Peer et al., 2017). Most specifically, Clickworker has previously been employed in cartographic and geovisualization research, demonstrating its suitability for web-based user experiments (Traun et al., 2021). A total 401 responses were collected before data cleaning. To improve data quality and reduce the influence of careless or unrealistic submissions, several quality-control procedures were applied. Incomplete responses, unrealistically short completion times, and suspicious response patterns were removed from the dataset. In addition, unusually long duration responses were manually inspected to distinguish inactivity from careful participation. After every experimental task, subjects had to rate the difficulty of the task to complete. The subjective difficult assessment was added to evaluate the cognitive workload and the ease in which the thematic map was understood, for each visualization method.

These responses were then compared to objective performance measures like interpretation accuracy and reaction time.

.Consistency in responses, response behaviour on tasks, and quality of answers were also taken into account when cleaning. After the cleaning, there were 178 valid responses in the three experimental groups. The DD group had 58 participants; the FI group had 62 participants and the UI group 58 participants. Balanced gender distribution was observed in the different experimental groups for the last sample. There were 29 male and 29 female participants in the DD group, 32 male and 30 female participants in the FI group, and 37 male and 21 female participants in the UI group. For distribution of age and gender by group see figure 6, for self-reported cartographic experience and education level see Table 1 and 2.

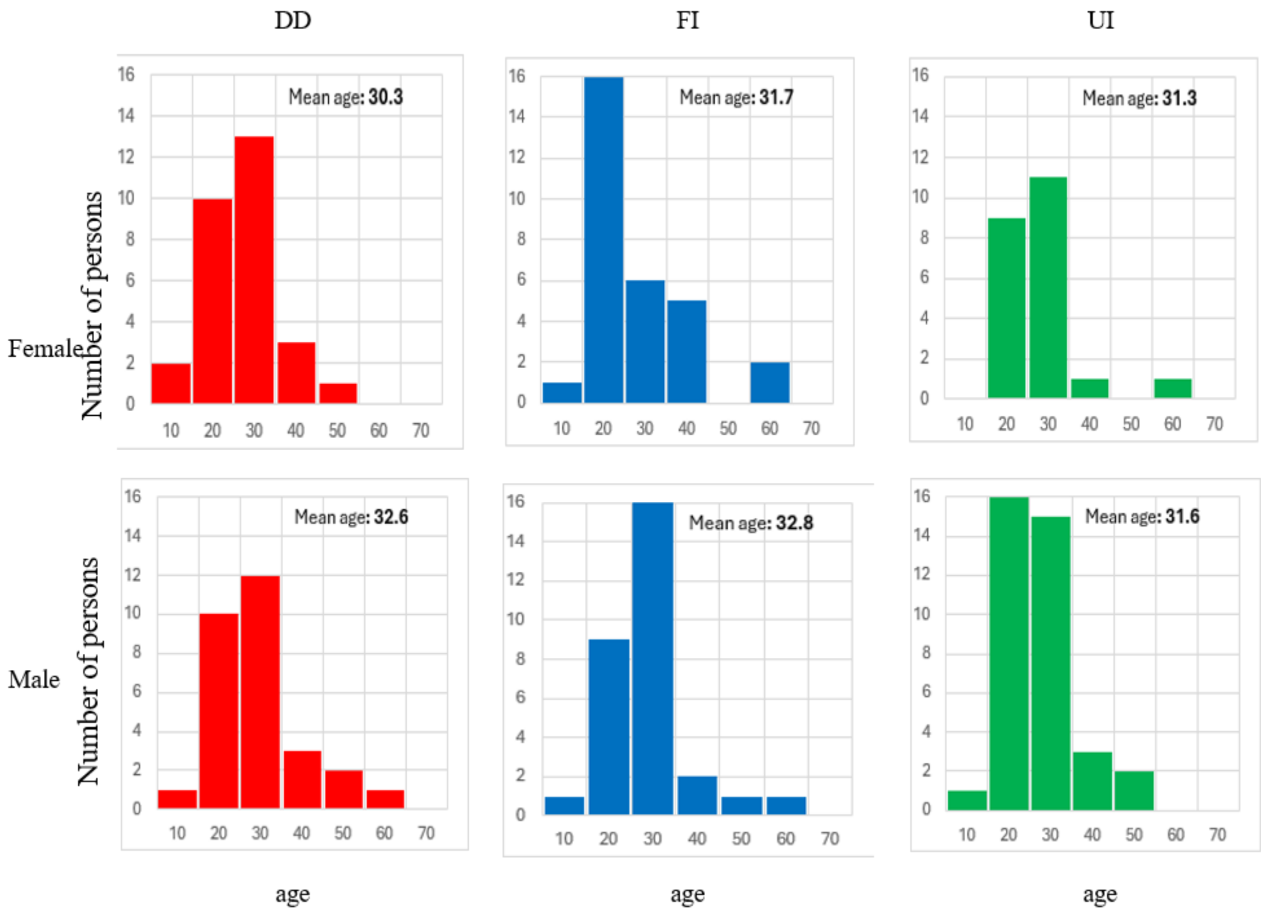


Figure 6: Age distribution and group

Table 1. Participants educational level (Raw count)

	DD	FI	UI	∑
High School	9	10	4	23
Bachelor's Degree	27	31	30	88
Master's Degree	17	15	15	47
PhD / Doctorate	3	4	4	11
Others	2	2	5	9
∑	**58**	**62**	**58**	**178**

Table 2: Self-reported cartographic experience (Raw count)

	DD	FI	UI	∑
Beginner	15	7	9	31
Basic	5	5	5	15
Intermediate	15	23	11	49
Advanced	13	15	24	52
Expert	10	12	9	31
∑	**58**	**62**	**58**	**178**

2.5 Data Analysis

Preprocessing and thematic map generation were done in ArcGIS Pro/Python GIS and spatial analysis software. Data collected were cleaned using quantitative statistical methods in Excel software to assess the difference in the performance of interpretation of the thematic map for the three visualization techniques. Participants' demographics, accuracy of interpretation, response time and difficulty ratings were summarized using descriptive statistics. To assess how well Dot Density maps, Filled Isopleths, and Unfilled Isopleths perform, they were compared with each other in a series of statistical analyses. Relationships between the subjective difficulty ratings and objective performance measures also were analyzed. The data was analysed statistically using Jamovi, a free and open-source statistical software application developed using R programming language. It was chosen due to its combination of the power and reproducibility of R with its user-friendly GUI. This software is able to perform various statistical procedures; thus, it can be used in quantitative research and academic uses (Navarro & Foxcroft, 2019; Sahin & Aybek, 2020; Silva de Souza et., 2024; Ashour, 2024).

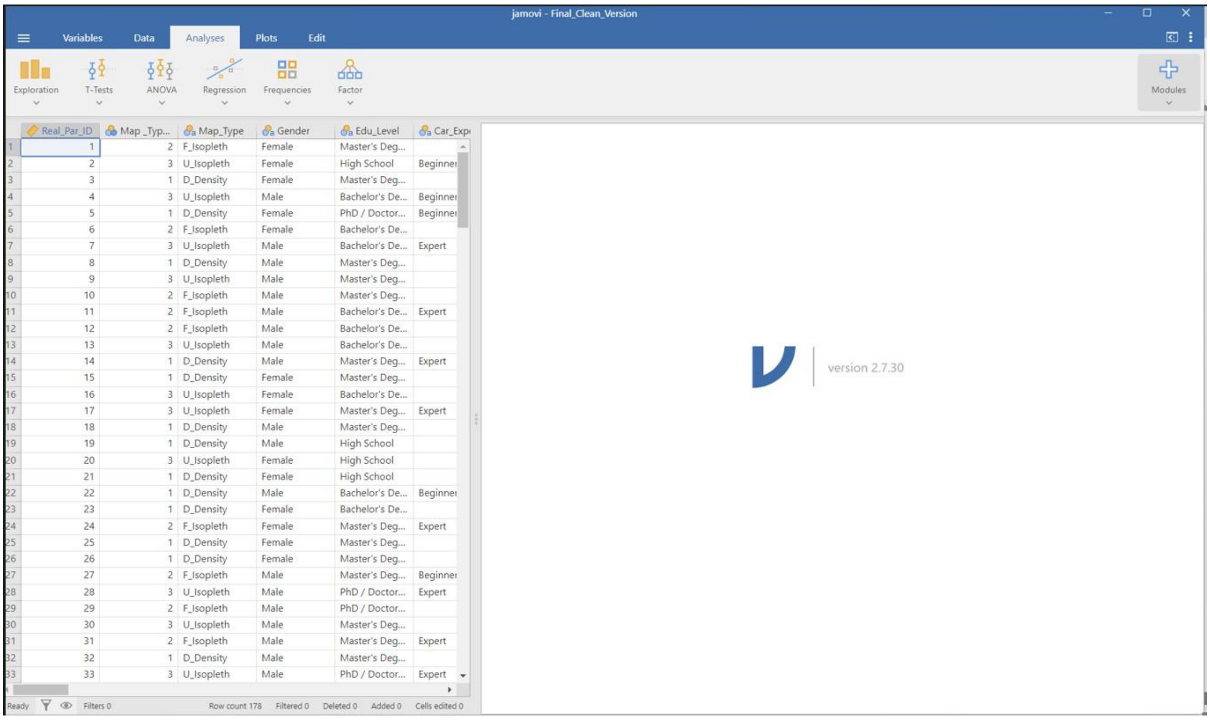


Figure 7: Jamovi (Version 2.7.30) interface displaying the main data view panel.

3. Result

3.1 Accuracy

3.1.1 Descriptive findings

The overall accuracy scores of the three thematic mapping techniques are shown in figure 8 for real-world and synthetic spatial interpretation tasks. The findings show that there are significant differences in performance between the spatial task and the cartographic representation employed. The DD representation gave the best average accuracy for real-world hotspot task (RW_HS) of the three maps. It indicates that in realistic spatial distributions, discrete point-based representation can be useful for hotspot localization because it enables the participant to easily identify the dense parts of the distribution in the visual domain. On this task, FI yielded less accuracy scores.

The real-world dataset (RW_GR) performance was better when using the gradient interpretation task for FI. The accuracy values of average accuracy were highest for FI, and comparatively low for DD and UI. The same pattern was seen for the synthetic hotspot tasks (SYN_HS_A) and SYN_HS_B) with FI continuing to perform best across both tasks. In case of the synthetic gradient tasks SYN_GD_A, FI and DD obtained nearly the same accuracy and UI was significantly lower. The differences between the types of maps were not so great in the SYN_GD_B, and the general performance was more uniform across the three representations.

In general, the outcomes suggest that the use of thematic mapping techniques has different effectiveness in relation to the type of spatial information task applied. DD maps have been found to be effective in identifying hotspots in some real-world applications, but in the cases of the synthetic pattern tasks, and gradient interpretation, performance of DD maps was generally lower than that of FI. The results confirm the notion that there is no single "best" thematic representation for all spatial interpretations.

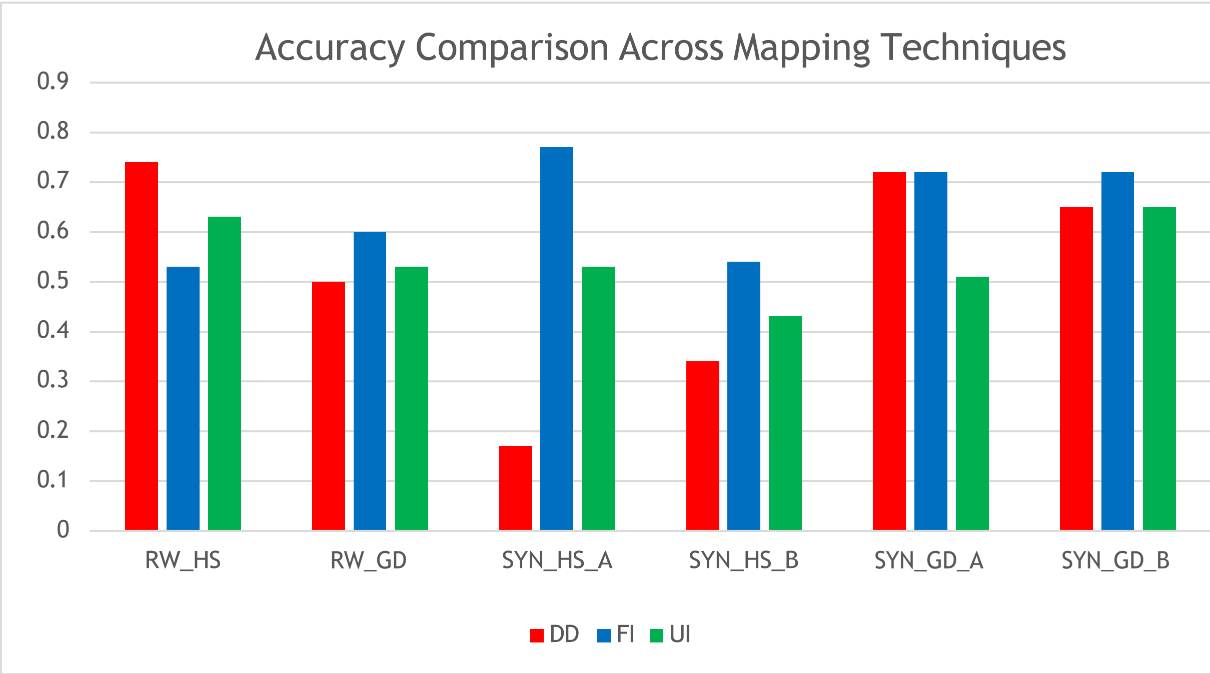
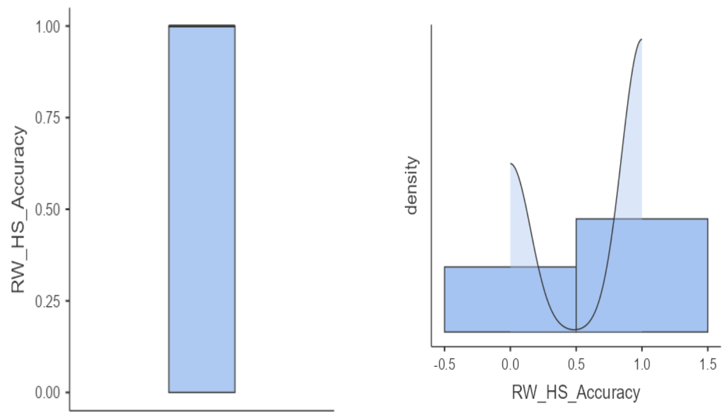


Figure 8: Average accuracy scores across thematic mapping techniques for real-world and synthetic spatial interpretation tasks.

3.1.2 Statistical Significance

Before doing inferential analysis, the properties of the data distribution were examined with a histogram, q-q plots, skewness statistics and the Shapiro-Wilk normality test. The results showed that there were significant deviations from normality for many of the variables.

Accuracy data consisted of binary categories outcomes (correct (1) versus incorrect (0) responses). Therefore, Chi-square tests of independence were applied to examine whether answer accuracy differed significantly across the thematic mapping techniques. Cramer's V was additionally calculated to assess the strength of association between map types and answer accuracy.



Descriptives

	RW_HS_Accuracy
N	178
Missing	0
Mean	0.635
Median	1.00
Standard deviation	0.483
Minimum	0
Maximum	1
Skewness	−0.565
Std. error skewness	0.182
Kurtosis	−1.70
Std. error kurtosis	0.362
Shapiro-Wilk W	0.609
Shapiro-Wilk p	< .001

Figure 9: Normality test Accuracy outcome of one of the Variables

The average accuracy for all the 6 tasks was 58%. Filled Isopleths achieved the highest overall accuracy 65%, outperforming both Dot Density maps 52% and Unfilled Isopleths 55%. The results suggest that participants were more successful when interpreting continuous filled density surfaces compare to discrete or contour-based representation.

The accuracy of the answers differed significantly between users of the three different map types: $X^2(2, N = 1068) = 13.3$, $p = 0.001$, Cramer's V = 0.112. Pairwise comparisons showed that the differences in answer accuracy between each pair of the tested map types were statistically significant significantly:

- DD-FI X^2 (1, N = 720) = 12.1, $p < .001$, Cramer's V = 0.130 (Filled Isopleth approximately 13 percentage points higher than Dot Density)
- DD-UI X^2 (1, N = 696) = 0.578, $p = 0.447$, Cramer's V = 0.29. (No statistically significant difference in answer accuracy was observed between Dot Density and Unfilled Isopleth)
- FI-UI X^2 (1, N = 720) = 7.33, $p = 0.007$, Cramer's V = 0.101 (Filled Isopleth maps producing approximately 11 percentage points higher accuracy than Unfilled Isopleth).

To account for repeated observations from the same participant and to further examine the influence of map type and spatial task type on interpretation accuracy, a logistic generalized linear mixed model (GLMM) was fitted with participant ID included as a random intercept. The model revealed a significant effect of map type, $X^2(2) = 7.45$, $p = 0.024$, indicating that accuracy differed across the thematic map representations. A significant effect of task type was also observed, $X^2(1) = 13.31$, $p < 0.001$, suggesting that participant performance varied between hotspot identification and gradient interpretation tasks. Furthermore, a significant interaction between map type and task type was found, $X^2(2) = 6.72$, $p = 0.035$, indicating that the effectiveness of the map types depended on the type of task being performed.
The GLMM indicated that FI representation produced significantly higher accuracy than DD representation ($p = 0.009$), whereas no significant difference was observed between the UI and DD representations ($p = 0.564$). these findings are consistent with the pairwise Chi-square analyses, which showed that the primary difference occurred between FI and DD.

Accuracy of answers differed also between tasks. See figure 10 for average percentage for each task.

The inferential statistical analysis as shown in Table 3 that answer accuracy differed across the tested thematic mapping techniques for several tasks. Chi-square tests of independence were conducted to examine whether the type of thematic map significantly influenced participants' interpretation accuracy. Cramer's V was additionally used to assess the strength of association between map type and task performance. Where statistically significant differences were identified, post-hoc pairwise comparisons were conducted to determine which mapping techniques differed significantly from one another. Statistical significance was evaluated at the 0.05 significance level.

For the Real-world Hotspot task (RW_HS), a statistically significant association was observed between map type and answer accuracy, $p = 0.059$, Cramer's V = 0.167. Pairwise comparisons showed that DD:FI ($p = 0.018$) However, no statistically significant difference was found between DD:UI and FI:UI.

Similarly, the Real-World Gradient task (RW_GD) showed no statistically significant relationship between map type and answer accuracy, $p = 0.557$.

For the Synthetic Hotspot A task (SYN_HS_A), a high statistically significant difference in answer accuracy was found among the three map types, $p = < 0.001$, Likewise pairwise comparisons reached statistical significance, Filled Isopleth outperformed both Dot Density and Unfilled Isopleth. DD:FI ($p = < 0.001$), FI:UI ($p = 0.006$). Unfilled Isopleth also received higher statistically significant over Dot Density ($p = < 0.001$)

In contrast, the Synthetic Hotspot B task (SYN_HS_B) shows no statistically significant association between map type and answer accuracy, $p = 0.079$, Pairwise comparisons indicated that Filled Isopleth maps achieved significantly higher accuracy than Dot Density maps (DD:FI $p = 0.025$).

For the Synthetic Gradient A task (SYN_GD_A), Shows statistically significant association between map type and answer accuracy, $p = 0.024$, Pairwise comparisons demonstrated that the success rates were nearly identical for DD:FI $p = 0.984$, therefore, there was not statistically significant. Both Dot Density and Filled Isopleth maps outperformed Unfilled Isopleth maps DD:UI ($p = 0.022$) and FI:UI ($p = 0.018$).

Similarly, the Synthetic Gradient B task (SYN_GD_B) showed no statistically significant association between map type and answer accuracy, $p = 0.629$.

Overall, the findings consistently indicate that Filled Isopleth maps supported higher interpretation accuracy across several hotspot and gradient interpretation tasks. Although the observed effect sizes ranged from small to moderate, the repeated statistical significance across multiple tasks suggests that Filled Isopleths provide a more effective representation for interpreting continuous population density patterns compared to Dot Density and Unfilled Isopleth techniques.

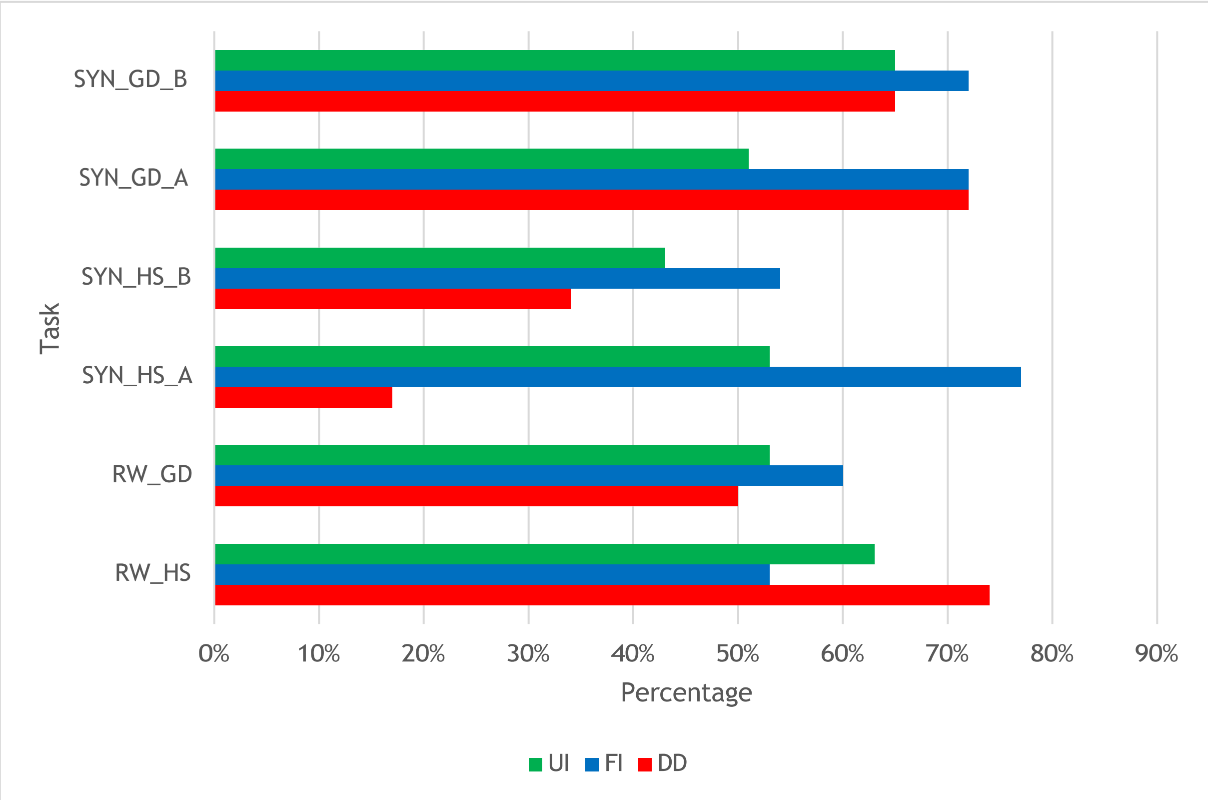


Figure 10: Percentage accuracy scores across thematic mapping techniques for real-world and synthetic spatial interpretation tasks.

Table 3. Inferential Statistical analysis of answer accuracy across the three thematic map types.

Task	Chi-Square Overall χ^2	*P*	Cramér's V	Pairwise Comparison	χ^2	Cramér's V	*P*	*Significant*
RW_HS	5.66	0.059	0.178	DD-FI	5.64	0.217	0.018	Yes
				DD-UI	1.45	0.112	0.229	---------
				FI-UI	1.38	0.107	0.241	---------
RW_GD	1.17	0.557	0.811	DD-FI	0.13	0.0972	0.136	---------
				DD-UI	0.138	0.0345	0.710	---------
				FI-UI	0.477	0.0628	0.491	---------
SYN_HS_A	43.8	< .001	0.496	DD-FI	43.5	0.602	< .001	Yes
				DD-UI	0.853	0.379	< .001	Yes
				FI-UI	3.09	0.253	0.006	Yes
SYN_HS_B	5.09	0.079	0.169	DD-FI	5.02	0.204	0.025	Yes
				DD-UI	0.908	0.0885	0.341	---------
				FI-UI	1.65	0.117	0.199	---------
SYN_GD_A	7.49	0.024	0.205	DD-FI	0.000	0.002	0.984	---------
				DD-UI	5.27	0.213	0.022	Yes
				FI-UI	5.56	0.215	0.018	Yes
SYN_GD_B	0.926	0.629	0.0712	DD-FI	0.701	0.0764	0.402	---------
				DD-UI	0.00	0.00	1.00	---------
				FI-UI	0.701	0.0764	0.402	---------

3.2Response time

3.2.1 Descriptive Findings

The average response times are shown in Figure 11 for real-world and synthetic interpretation tasks, across the three thematic mapping techniques. Results show significant differences in interpreting speed on the spatial task and on the cartographic representation used.

The response times were relatively low for the real-world hotspot task (RW_HS) for all three mapping techniques, suggesting that participants could identify the hotspots quickly and DD had the lowest response time while interpretation times were slightly longer for FI and UI.

The real-world gradient task (RW_GD) yielded significantly more response time for all mapping techniques in comparison. This indicates that identifying a hotspot would be easier than interpreting the gradient. There was also not a large difference between DD, FI and UI on this task, suggesting that all three representations required similar processing time for real world interpretation of gradients.

For the synthetic hotspot tasks, significant differences were found between the mapping techniques, with differences showing longer response times when participants performed UI than DD and FI, which may indicate that they needed more time to interpret the hotspot pattern on the synthetic map. The same tendencies were noted in the case of SYN_HS_B, but they were less pronounced than those noted in the case of SYN_HS_B.

The synthetic gradient tasks (SYN_GD_A and SYN_GD_B) also revealed increased interpretation times, particularly for UI and FI. FI recorded the fastest response time in SYN_GD_A and the longest interpretation time in SYN_GD_B when compared to the other mapping techniques. The present results suggest that continuous and contour-based representations might need more detailed visual processing in order to interpret gradual synthetic density transitions.

Overall, the response time analysis suggests that hotspot tasks were generally interpreted faster than gradient task across all mapping techniques. While DD representations frequently supported faster task completion, FI and UI often required longer interpretation times, particularly in synthetic gradient scenarios. The results therefore indicate that interpretation efficiency varies depending on both spatial patterns being represented and the cartographic techniques used.

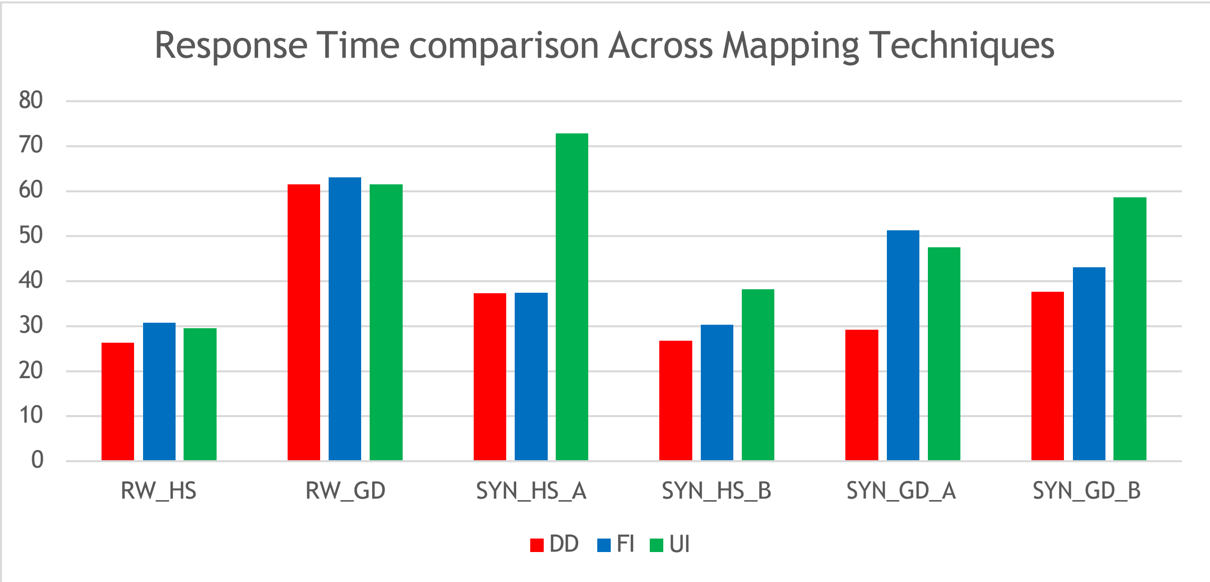


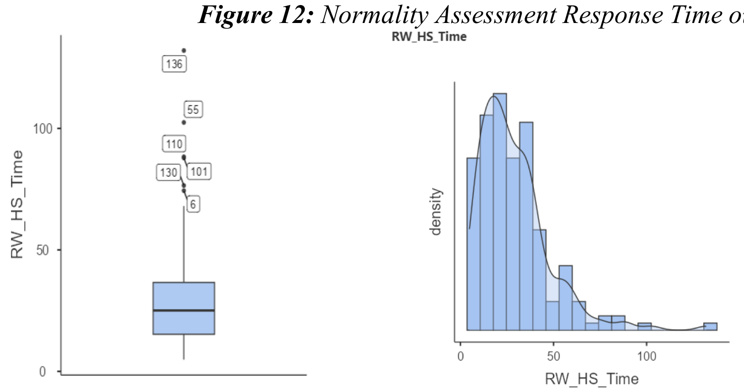
Figure 11: Average response time across thematic mapping techniques for real-world and synthetic spatial interpretation tasks

3.2.2 Statistical Significance

Response-time variables demonstrated positively skewed distribution and did not satisfy the assumptions of normality. Similarly, perceived difficulty ratings were measured using Likert-scale responses and showed non-normal distributions. Consequently, non-parametric Kruskal-Wallis tests were performed to compare differences in response time and perceived difficulty across the thematic mapping techniques.

Where statistically significant differences were identified between map types, post-hoc pairwise comparisons were conducted to determine which mapping techniques differed significantly from one another. Statistical significance was evaluated at the 0.05 significance level.

Figure 12: *Normality Assessment Response Time outcome of one of the Variables*



Descriptives

	RW_HS_Time
N	178
Missing	0
Mean	28.9
Median	25.1
Standard deviation	19.1
Minimum	4.85
Maximum	132
Skewness	1.84
Std. error skewness	0.182
Kurtosis	5.64
Std. error kurtosis	0.362
Shapiro-Wilk W	0.862
Shapiro-Wilk p	< .001

A Kruskal-Wallis's test was conducted to examine differences in response time across the three thematic mapping techniques. The analysis revealed a statistically significant difference in response time between map types, $X^{2(}2) = 8.23$, $p = 0.016$, with a small effect size (e = 0.008).

Post hoc Dunn's pairwise comparisons with Bonferroni correction showed a statistically significant difference between DD:FI maps ($p = 0.022$). However, no statistically significant differences were observed between FI:UI ($p = 1.000$) and DD:UI maps ($p = 0.077$). The test showed no statistically significant difference in response time between the three map types for the RW_HS task, $p = 0.309$. A statistically significant difference in response time between the three map types was revealed in RW_GD task ($p = 0.030$), the pairwise comparison are as follows FI:UI ($p = 0.033$), and there are not statistically significant between DD:FI $p = 1.000$ or DD:UI ($p = 0.165$).

For SYN_GD_A task $p = 0.002$, DD:FI ($p = 0.012$), DD:UI ($p = 0.005$). however, no statistically significant difference was observed between FI:UI ($p = 1.000$). there was not statistically significant between the map types for SYN_GD_B task, $p = 0.82$.

A statistically significant difference in response time between the three map types for SYN_HS_A was revealed $p = 0.036$, the pairwise comparison shows a statistically significant difference in FI:UI (p= 0.031). No statistically significant difference was observed in DD:FI ($p = 0.937$) or DD:UI ($p = 0.376$). for the SYN_HS_B task $p = 0.340$.

Table 4. Inferential Statistical analysis of response time across the three thematic map types.

Task	*Kruskal Wallis H*	*P*	Map Type	M (s)	SD	Post Hoc Groups	*p*
RW_HS	2.35	0.309	DD	26.3	15.8	-------	Not Statistically Significant
			FI	30.72	17.9		
			UI	29.5	22.8		
RW_GD	7.00	0.030	DD	61.5	46.0	---------	------------
			FI	63.0	39.3	---------	------------
			UI	61.5	81.3	FI-UI	0.033
SYN_GD_A	12.10	0.002	DD	29.2	14.6	DD-FI	0.012
			FI	51.3	77.9	DD-UI	0.005
			UI	47.5	33.7	---------	-----------
SYN_GD_B	4.99	0.082	DD	37.7	25.8	--------	Not Statistically Significant
			FI	43.0	21.2		
			UI	58.6	67.5		
SYN_HS_A	6.67	0.036	DD	37.3	28.9	---------	---------
			FI	37.5	39.7	---------	---------
			UI	72.7	137	FI-UI	0.031
SYN_HS_B	2.16	0.340	DD	26.8	19.6		Not statistically Significant
			FI	30.4	18.3		
			UI	38.2	41.7		

3.3 Perceived Difficulty Analysis

3.3.1 Descriptive Findings

The results shown in Figure 13 indicate that the average perceived difficulty ratings across the three thematic mapping techniques for both real-world and synthetic interpretation tasks. Overall, the results demonstrate noticeable differences in subjective task difficulty depending on both the cartographic representations and the spatial interpretation task. For the real-world hotspot task (RW_HS), DD received the lowest difficulty rating among the three mapping techniques, indicating that participants perceived hotspot identification using point-based representations as relatively easy. In contrast, FI and UI produced slightly higher difficulty ratings for this task.

The real-world gradient task (RW_GD) generated higher perceived difficulty values across all mapping techniques compared to hotspot tasks. UI showed the highest difficulty rating, suggesting that contour-based representations may require greater cognitive effort when interpreting gradual density transitions. DD and FI demonstrated similar difficulty levels in this task.

For the synthetic hotspot tasks, FI consistently produced the lowest perceived difficulty values, particularly in SYN_HS_A. This suggests that continuous surface representation may support easier interpretation of synthetic clustered spatial patterns. In contrast, UI generally produced higher difficulty ratings, especially in SYN_HS_B. The synthetic gradient tasks (SYN_GD_A and SYN_GD_B) also revealed comparatively elevated difficulty ratings across all mapping techniques. UI produced the highest difficulty values in several gradient scenarios, while FI often demonstrated slightly lower perceived difficulty. These findings indicate that contour-based representations may increase interpretation complexity when participants attempt to identify gradual synthetic density variations.

Overall, the perceived difficulty analysis suggests that hotspot tasks were generally considered easier than gradient interpretation tasks across all mapping techniques. In addition, FI frequently demonstrated lower perceived difficulty in synthetic interpretation tasks, whereas UI often required greater cognitive effort. The results further highlight the relationship between cartographic representation, task complexity, and subjective user experience during spatial density interpretation.

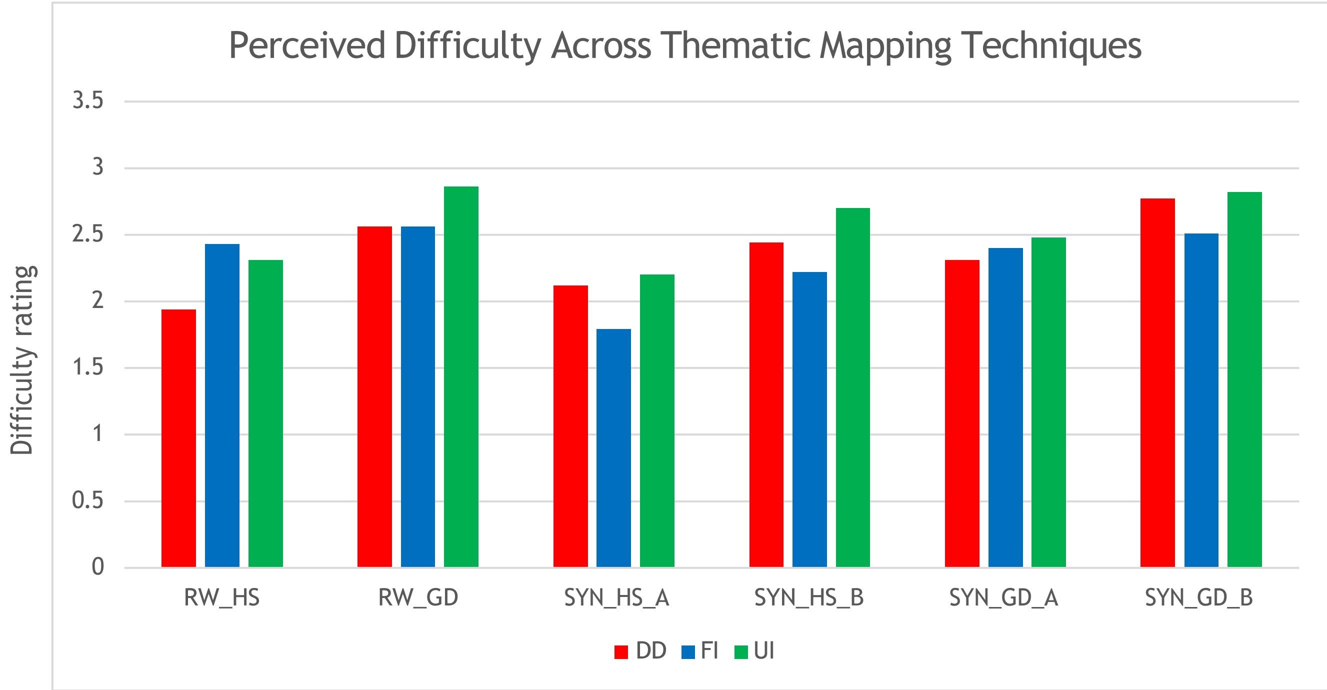


Figure 13: Average difficulty rating across thematic mapping techniques for real-world and synthetic spatial interpretation tasks

3.3.2 Statistical Significance

For the overall perceived task difficulty, A Kruskal-Wallis's test revealed a significant effect of the map on perceived task difficulty, $X^2(2) = 12.2$, $p = 0.002$. participants rated Filled Isopleth maps as the easiest (M = 2.32, Median = 2), followed by Dot Density maps (M = 2.36, Median = 2). Unfilled Isopleth received the highest difficulty ratings (M = 2.57, Median = 3). Dunn's post hoc comparisons with Bonferroni correction showed that Unfilled Isopleth maps were perceived as significantly more difficult than both Dot Density maps (p = 0.030) and Filled Isopleth maps (p = 0.003). No significant difference in perceived difficulty was observed between Dot Density and Filled Isopleth maps (p = 1.000).

RW_HS task shows a statistically significant difference in perceived difficulty among the three map types $p = 0.035$. However, only DD:FI $p = 0.026$ indicated a significant difference and does not exist between DD:UI $p = 0.226$ and FI:UI $p = 1.000$. No statistically significant difference among the three map types for the RW_GD task $p = 0.218$. therefore, participants perceived the difficulty of the task similarly across DD, FI, and UI.

A significant difference in perceived difficulty was revealed in SYN_HS_A task, $p = 0.023$. FI:UI $p = 0.029$ and no significant difference were found between DD:FI $p = 0.112$ and DD:UI p = 1.000. Also, in SYN_HS_B, a significant difference was revealed among the three map types $p = 0.027$. only FI:UI $p = 0.022$ shows significant difference and none exist between DD:FI $p = 0.684$ and DD:UI $p = 0.439$.

Perceived difficulty did not differ significantly among the three map types for the SYN_GD_A task $p = 0.674$. consequently, not statistically significant pairwise differences were observed between any of the map types in the post hoc Dunn' tests (all Bonferroni-adjusted $p = 1.000$). Similarly, no significant difference in SYN_GD_B task $p = 0.119$. none of the pairwise comparisons between map types were statistically significant (all Bonferroni adjusted $p > 0.05$).

Table 5. Inferential Statistical analysis of perceived difficulty rating across the three thematic map types.

Task	**Kruskal-Wallis's x^2**	***p***	**Significant Pairwise Comparison**	***p***
RW_HS	8.37	0.015	DD:FI	0.026
RW_GRD	3.05	0.218	- - - - - - - - - - - -	- - - - - - - - -
SYN_HS_A	7.58	0.023	FI:UI	0.029
SYN_HS_B	7.20	0.027	FI:UI	0.022
SYN_GRD_A	0.79	0.674	- - - - - - - - - - - -	- - - - - - - - -
SYN_GRD_B	4.25	0.119	- - - - - - - - - - - -	- - - - - - - - -

3.4 Comparative Analysis of Mapping Techniques

3.4.1 Accuracy, Response Time, and Perceived Difficulty Trade-offs

The comparative analysis reveals important relationships between interpretation accuracy, response time, and perceived difficulty across the three thematic techniques. Overall, the results indicate that no single representation consistently outperformed the others across all spatial interpretation tasks. Rather, the performance of the mapping methods varied greatly according to the type of the spatial pattern being explained.

Across many of the tasks, such as the gradient examples, and a number of the synthetic scenarios, Filled Isopleths (FI) often provided the most accurate results. In SYN_HS_A, SYN_HS_B, and the two synthetic gradient tasks, for instance, FI consistently performed best versus DD and UI. While such higher accuracy values often came with comparatively longer response times, particularly in SYN_GD_A and RW_GD. This indicates that it may have taken the participants longer to interpret continuous density surfaces correctly.

In general, perceived difficulty ratings for FI were moderate, and in some synthetic tasks, lower than for DD and UI, although longer response times were observed with FI in some tasks. This could suggest that while interpreting times were slightly longer, participants found that FI were cognitively manageable, visually supportive for interpreting spatial transitions, and facilitated the understanding of the gradual transitions.

Dot Density (DD), however, often yielded the lowest response times for various tasks such as in RW_HS, SYN_HS_B, SYN_GD_A and SYN_GD_B. This indicates that point-based representations might facilitate faster visual scanning and speed up decision making. Furthermore, DD also outperformed the other algorithms in the real-world hotspot task (RW_HS) indicating that discrete point clustering is effective for hotspot localization in realistic spatial environments. The decrease in the DD performance was found to be significant in synthetic hotspot and gradient tasks especially in SYN_HS_A and SYN_HS_B, where FI showed better accuracy. The results indicate that, although DD can be used to interpret information quickly, it may not be suitable for the interpretation of continuous density transitions or controlled synthetic spatial structure.

Comparatively higher response times and higher perceived difficulty ratings were found across a few tasks for Unfilled Isopleths (UI), overall. Characterized by longer responses times in SYN_HS_A and SYN_GD_B, UI also exhibited high perceived difficulty scores in certain gradient tasks. The patterns imply that contours might be more cognitively challenging for participants, particularly in the case of complex synthetic gradients and clustered patterns. The contour interpretation was capable of moderate accuracy in some cases, but the overall performance indicates that this task might be more time consuming than DD and FI.

Combined, the findings suggest important quantifiable trade-offs between the efficiency, accuracy, and subjective effort of interpretation. The faster the interpretation the less accurate it was, and more difficult representations were not necessarily perceived as slow. The results point to the intricate relationship between the representational level of the map, spatial task type and human visual interpretation behaviour.

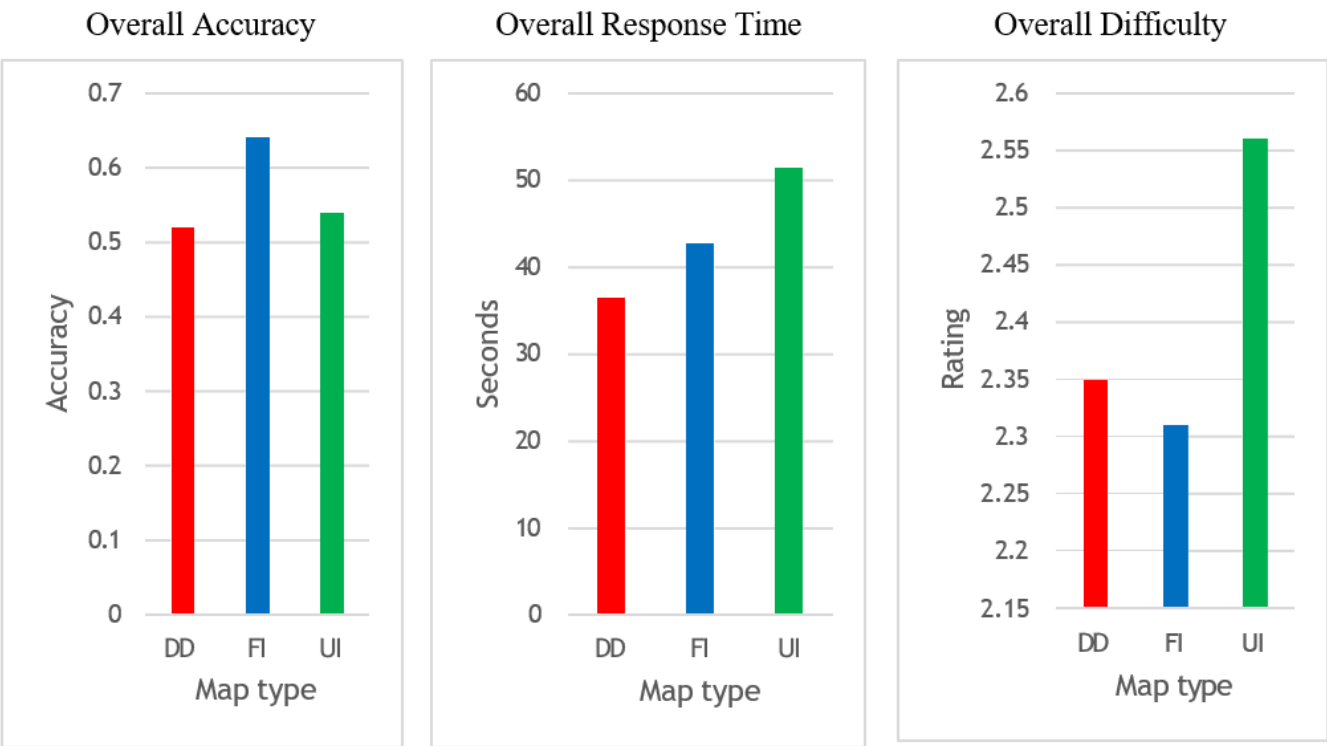


Figure 14: Comparative trade-offs between interpretation accuracy, response time, and perceived difficulty across thematic mapping techniques.

3.4.2 Hotspot VS Gradient Interpretation

3.4.2.1 Descriptive findings

The comparison between hotspot and gradient interpretation tasks showed that there were significant differences between the two in terms of the participants' performance and cognitive processing behavior. In all three thematic mapping methods, the response time and difficulty ratings were lower for the hotspot tasks compared to the gradient interpretation tasks. It seems that people can detect a dense spatial gradient better than they can detect a diffuse one.

For the real-world hotspot task (RW_HS) good accuracy and low response times were observed using all mapping techniques, especially DD. This means that discrete point-based representations could be useful for intuitive recognition of localised concentration patterns in realistic spatial environments. Participants also found hotspot identification tasks easier than gradient interpretation tasks, which also confirmed the hypothesis that hotspot identification is easier because of the clustering effect.

The response times and perceived difficulty ratings were consistently higher for gradient interpretation tasks, however. The gradual spatial transition interpretation task was particularly apparent in RW_GD, SYN_GD_A and SYN_GD_B, as these tasks took longer for participants to complete. The higher cognitive demand of the gradients is perhaps due to the need to

Continuously assess variations in density horizontally, not just in hotspots.

Filled isopleths performed very well in the task of interpreting gradients. For the continuous surface representations, the accuracy values were consistently higher in RW_GD, SYN_GD_A and SYN_GD_B, indicating that the continuous surfaces were successful in communicating the gradual spatial variation. Smooth density transitions, however, were found to be less well-represented by discrete point distributions as seen in the lower accuracy observed in gradient situations for DD.

The results support the notion of the task-dependency of thematic mapping effectiveness. FI could be more effective in gradient interpretation and continuous density analysis while DD is very effective for location of hotspot.

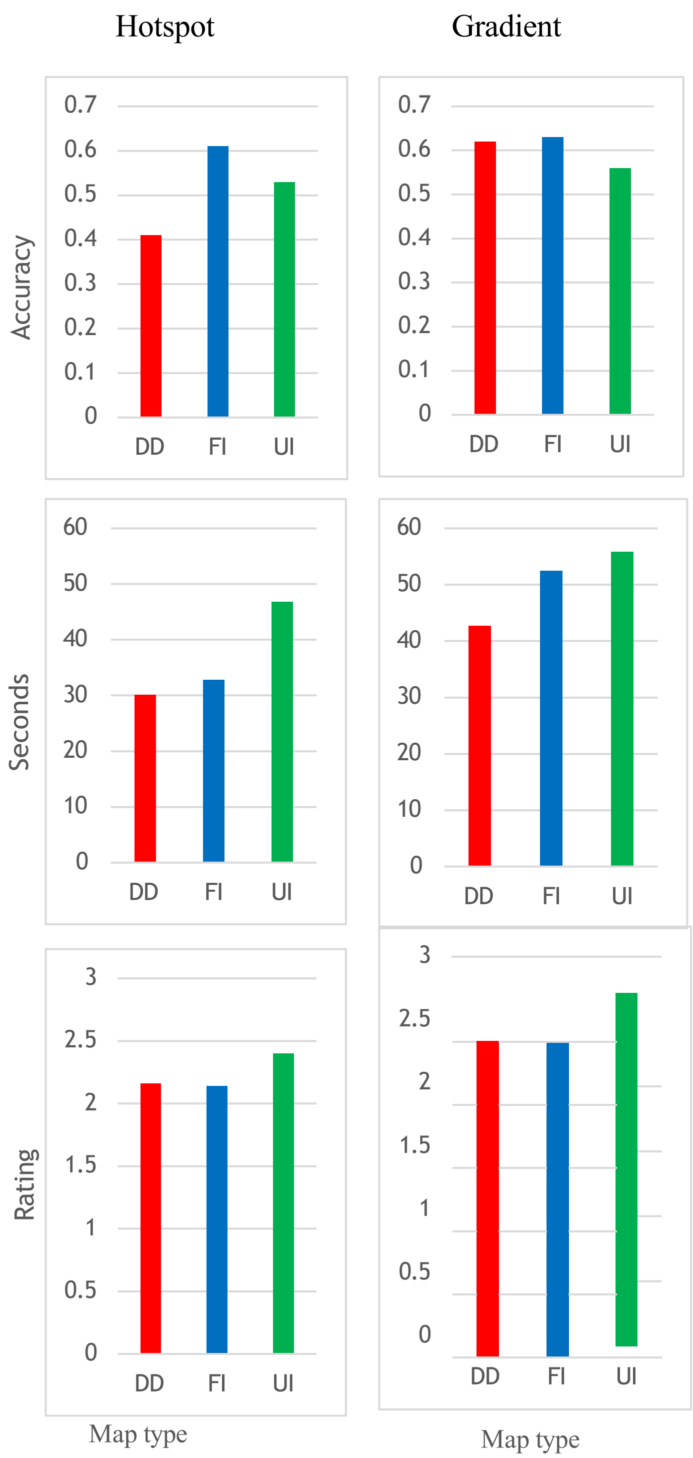


Figure 15: Comparative analysis of hotspot and gradient interpretation tasks across thematic mapping techniques. The figure summarizes differences in accuracy, response time, and perceived difficulty for DD, FI, and UI representation across real-world and synthetic spatial interpretation tasks.

3.4.2.2 Statistical Significance

There was a significant difference in accuracy for the three thematic map types in the hotspot identification tasks (X2(2, N = 534) = 14.3, $p < 0.001$, Cramer's V = 0.164). FI achieved the highest accuracy (61.8%), followed by UI (53.4%) and DD (42.0%). The study indicates that continuous filled surface representation was most effective for identifying hotspots than the other two representations, contour and point.

By contrast, no statistically significant difference in answers accuracy was found between the three thematic maps for the interpretation of the gradients, X2(2, N = 534) = 4.98, $p = 0.083$, Cramer's V = 0.097. While the accuracy for FI, DD and UI was highest at 68.3%, 62.6% and 56.9%, respectively, the differences between the three map types were not significant enough to suggest any meaningful effect when measuring their accuracy in the task of gradient interpretation.

Table 6: Chi-square test comparing answer accuracy across thematic map types for hotspot and identification and gradient interpretation tasks.

Spatial Task Type	**X**	***p*-value**	**Cramer's V**
Hotspot	14.30	< 0.001	0.164
Gradient	4.98	0.083	0.097

3.4.3 Real-World VS Synthetic Tasks

There were also significant differences between the real-world and the synthetic datasets in regard to interpretation behaviour and mapping performance in the comparison. Real world tasks were more complex in terms of spatial distribution and visual complexity compared to the synthetic datasets which were constructed with controlled spatial structures to isolate certain density patterns.

With real-world tasks, the differences among the mapping techniques were sometimes not as great. For instance, while the accuracy of DD was the highest for both RW_HS and RW_GD, FI outperformed for RW_GD, the difference in accuracy between techniques was relatively small. This may be a consequence of the natural variation and background noise in the actual distribution of people in space.

The synthetic tasks, on the other hand, often resulted in larger mapping technique performance differences. In SYN_HS_A, and the synthetic gradient tasks, FI showed particularly strong accuracy results, while DD had significantly lower accuracy in some of the synthetic tasks. The differences could be a result of controlled datasets that were generated and intentionally designed to highlight hotspot and gradient properties.

There were also larger response time and perceived difficulty differences for the synthetic tasks. In certain synthetic tasks, UI showed much faster response times and higher perceived difficulty ratings, indicating that contour-based representations could be more cognitively challenging in cases where the spatial pattern could be abstract and/or tightly controlled.

In general, the comparison of real world and synthetic datasets shows that the structure and spatial complexity of the datasets is of great impact to thematic map interpretation. Natural datasets have natural irregularities and contextual complexity, while synthetic datasets offer more experimental control and more significant differences in cartographic performance among different mapping techniques.

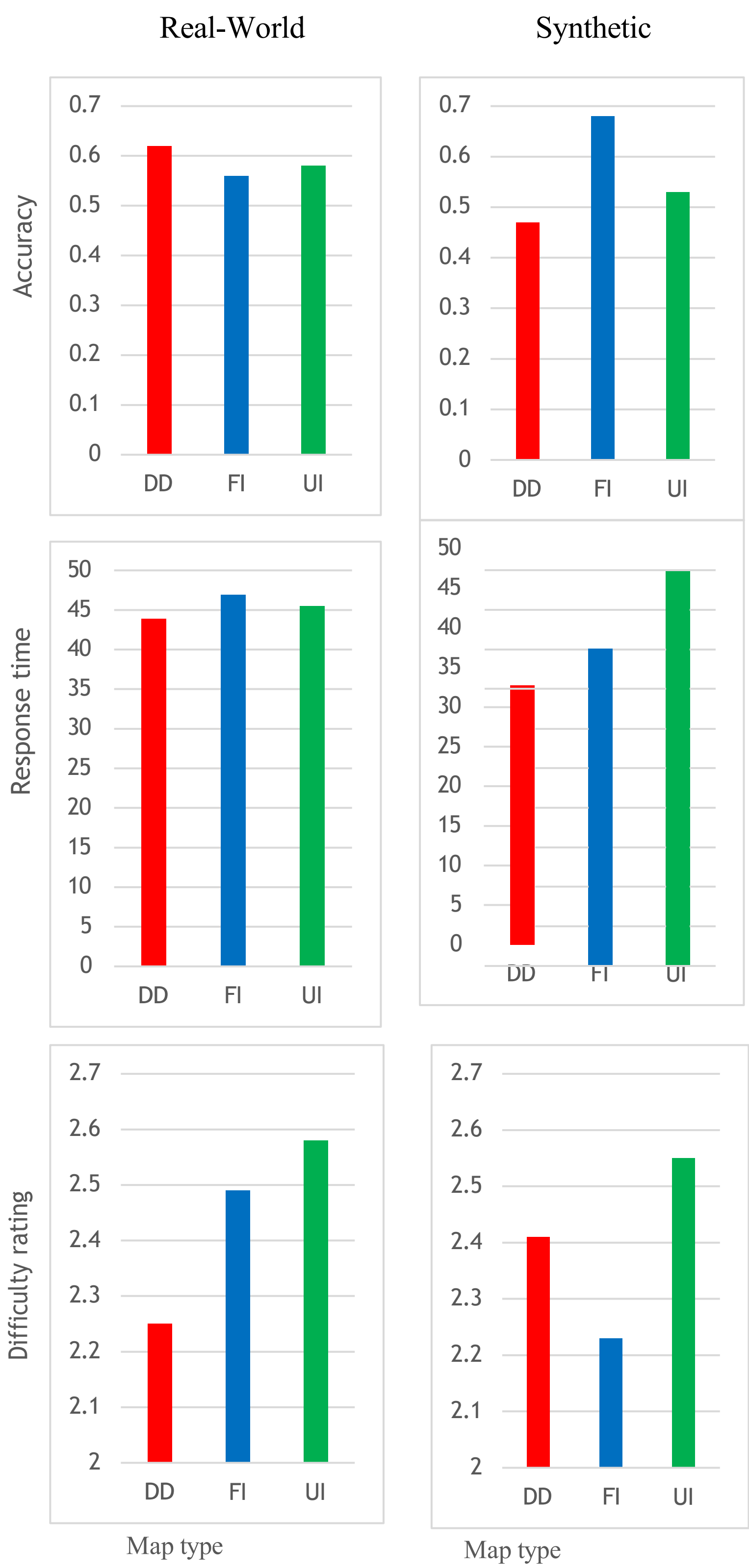


Figure 16: Comparison of interpretation performance between real-world and synthetic spatial tasks across thematic mapping techniques. Real-world tasks include RW_HS and RW_GD, while synthetic tasks include SYN_HS_A, SYN_HS_B, SYN_GD_A, and SYN_GD_B

3.5 Relationship Between Subjective Difficulty and Objective Performance

The extent to which the difficulty of the tasks was perceived was compared with accuracy and response time in the objective measure of interpretation performance. Task difficulty was rated on ordinal scales and accuracy was coded as a dichotomous variable (correct vs. incorrect), so both the correlation and group comparison methods were used. Spearman's rank correlation was used to investigate the relationship between perceived difficulty and performance measures; Mann-Whitney U tests were employed to determine whether there was a difference in perceived difficulty between correct and incorrect response categories. These analyses complement each other and offer complementary insights into the relationship between subjective perceptions of task difficulty and objective measures of task performance.

3.5.1 Difficulty vs Accuracy

For the real-world hotspot task (RW_HS), higher perceived difficulty was associated with lower accuracy (ρ =-0.155, $p = 0.039$). participants who answered incorrectly also reported significantly higher difficulty ratings than those who answered correctly (U = 2974, $p = 0.040$). in contrast, real world gradient task (RW_GD), no significant relationship was observed between perceived difficulty and accuracy (ρ = 0.044, $p = 0.562$). similarly, difficulty ratings did not differ significantly between participants who answered correctly and those who answered incorrectly (U = 3737, $p = 0.561$).

For the synthetic hotspot task (SYN_HS_A), participants who perceived the task more difficult tended to achieve lower accuracy scores (ρ = -0.226, $p = 0.02$). This finding was further supported by significantly higher difficulty ratings among participants who answered incorrectly (U = 2982, $p = 0.03$) and in SYN_HS_B, perceived difficulty showed a weak negative association with accuracy (ρ = -0.124, $p = 0.100$). However, neither the relationship nor the difference in difficulty ratings between correct and incorrect responses was statistically significant (U = 3372, $p = 0.100$). No significant relationship was observed in SYN_GD_A between perceived difficulty and accuracy (ρ = -0.002, $p = 0.976$). likewise, difficulty ratings did not differ significantly between participants who answered correctly and those who answered incorrectly (U = 3559, $p = 0.977$). Similarly, for SYN_GD_B, no significant relationship was observed between perceived difficulty and accuracy (ρ = 0.001, $p = 0.992$). also, difficulty ratings did not differ significantly between participants (U = 3446, $p = 0.994$).

Table 7: Results of correlation and group comparison analyses between perceived task difficulty and interpretation accuracy across experimental tasks

Task	**Spearman ρ** (Difficulty- Accuracy)	***p***	**Mann-Whiteney U**	***p***
RW_HS	-0.155	0.039	2974	0.040
RW_GD	0.044	0.562	3737	0.561
SYN_HS_A	-0.226	0.02	2982	0.03
SYB_HS_B	-0.124	0.100	3372	0.100
SYN_GD_A	-0.002	0.976	3559	0.977
SYN_GD_B	0.001	0.992	3446	0.994

The relationship between perceived difficulty and accuracy was generally weak across the experimental tasks. Significant negative associations were observed for the real-world hotspot and synthetic hotspot task, indicating that participants perceived these tasks as more difficult tended to achieve lower scores. The results for the gradient interpretation tasks showed no significant relationship between perceived difficulty and performance, that is, performance seemed to be relatively independent of perceived difficulty. Based on these results, it can be inferred that subjective difficulty could be more important than gradient interpretation tasks for identifying a hotspot, and that the effects observed were in general but not to an excessive degree.

3.5.2 Difficulty vs Response Time

In the real-world hotspot task, a weak positive correlation was found between perceived difficulty and response time ($\rho = 0.015$; $p = 0.038$). There was also a stronger positive correlation for the real-world gradient task ($\rho = 0.318$, $p < 0.001$), such that the more difficult a participant saw the task as, the longer it took them to complete. The same trend was found to occur with a synthetic hotspot task A ($\rho = 0.217$, $p = 0.004$). There were also positive correlations for synthetic hotspot task B ($\rho = 0.217$, $p = 0.072$), synthetic gradient task A ($\rho = 0.096$, $p = 0.204$), and synthetic gradient task B ($\rho = 0.133$, $p = 0.078$) but they were not statistically significant.

Table 8: Spearman correlation results between perceived task difficulty and response time across experimental tasks

Task	**Spearman ρ (Difficulty-Response time)**	***p***	***Interpretation***
RW_HS	0.156	0.038	Weak positive, significant
RW_GD	0.318	< 0.001	Moderate positive, significant
SYN_HS_A	0.217	0.004	Weak positive, significant
SYB_HS_B	0.135	0.072	Weak positive, not significant
SYN_GD_A	0.096	0.204	Very weak, not significant
SYN_GD_B	0.133	0.078	Weak, not significant

Overall, perceived difficulty was more consistently associated with response time than with accuracy. Significant positive relationships were observed for the real-world hotspot, and synthetic hotspot A, suggesting that participants tended to spend more time on tasks they perceived as difficult. In contrast, the relationship between perceived difficulty and accuracy was generally weak and only significant for two hotspots tasks. These findings suggest that subjective difficulty is more strongly reflected in task completion time than in interpretation accuracy.

3.6 Influence of Participants Characteristics

The characteristics of the participants are summarized in Table 1, Table 2 and Figure 6. Further exploratory analyses were performed to explore whether these characteristics impacted overall task performance. Overall accuracy was not statistically different by level of education, but was close to being significant by level of cartographic experience (Table 2). In addition, distributions of age (Figure 6) and gender were balanced and there were no significant differences among the experimental groups. In general, it seemed that the demographic and experiential factors had little impact on participant performance. Thus, these variables were not further explored in the primary analyses.

4. Discussion

4.1 Overall Performance of the Thematic Map Types

The results show that the design of the thematic maps had a strong effect on the participants' perception of population density patterns. The highest accuracy overall (65%) was obtained by Filled Isopleths, followed by Unfilled Isopleths (55%) and Dot Density maps (52%) for all tasks. Meanwhile, Dot Density maps tended to yield the least response time and unfilled Isopleths were often found with the highest difficulty ratings. The results suggest that the representation of a theme is different in its support of spatial analysis tasks. Filled Isopleths seem to be most useful for good interpretation. Filled Isopleths may be more accurate than other maps due to the continuous nature of the representation of population density. Continuous colour changes enable a user to know if the area is a high or low density and still see the gradual spatial transitions Shaito et al., (2022). Colour value is an effective visual variable because the human visual system is able to quickly identify light and dark elements, allowing to communicate spatial structure and ordered pattern, according to Zanin et al. (2013). In doing so, Filled Isopleths can help ease the cognitive load of understanding density variation, and may give the user a more direct and accurate sense of spatial variation than can be obtained from a point-based or contour-based representation.

This is in accordance with the empirical user study conducted by Słomska-Przech and Gołębiowska (2021) that showed the type of thematic map influenced the performance of the interpretation and that the maps that best captured overall patterns led to the greatest accuracy in interpretation. They also found that accuracy, response time, and difficulty of perception tended to follow the same pattern for different types of maps.

While overall accuracy was slightly lower compared to the other types of maps, Dot Density maps often enabled participants to make a response more rapidly, potentially indicating that they were making an approximation rather than processing the map in detail (Roth et al., 2019, 2020). Quick responses did not always result in more accurate interpretations. Overall, these results indicate that it is not possible to use a single performance measurement to describe the usability of thematic maps, since different representations can optimize different aspects of user performance.

4.2 Hotspot Identification Tasks

This study aimed at comparing the effectiveness of the plotting methods of Dot Density, Filled Isopleths and Unfilled Isopleths in identifying hot spots. The result indicate that Filled Isopleth was generally the best identifier of the hotspots in the population. In the case of hotspot identification, Filled Isopleths may have needed less cognitive effort to recognise concentrated population regions, which led to their superior performance. This result aligns with earlier user-centred cartographic studies demonstrating that the effectiveness of reading a map and the success of a task is influenced by its thematic map design. Research had shown that representations focusing on the general pattern of spatiality have yielded better understanding than representations involving estimation of values from individual symbols or more elaborate visual structures (Słomska-Przech et al., 2021).

There was also relatively good performance for hotspot identification of Dot Density maps. It is no surprise that Dot Density maps performed well given that these maps are particularly suited to visualizing local concentrations and spatial clustering. Based on the previous user evaluations, it can be concluded that dot maps have a high level of visual intuitiveness in relation to the ability to recognize the distribution of phenomena, but the accuracy of the interpretation might decrease if the user has to estimate quantitative differences between areas instead of identifying the clusters (Wielebski, & Medyńska-Gulij, 2023). In many cases, this loss of accuracy is due to a degree of discomfort experienced on viewing the touch and overlap of dots, a well-known limitation of this approach at high dot densities.

Unfilled Isopleths gave the lower accuracy and higher difficulty level for the Hotspot task. Raw contour lines, unlike Filled Isopleths, don't have a visible continuity and it's up to the user to monitor line spacing and to mentally interpolate densities in unshaded areas. This extra mental work may have resulted in a reduction in user performance, supporting the views of Peters and Meng (2015) that the unsegmented approach of merely line layering in map reading makes it a “difficult endeavour”. Overall, the results indicate that although all three map types can convey the location of the hot spots, Filled Isopleths are the most clear and easy-to-use map type for revealing areas of high population density.

4.3 Spatial Gradient Interpretation

These variations in maps were even more pronounced in the case of tasks involving gradient interpretation. Gradient interpretation is more complicated than hotspot identification because users must be aware of continuous changes in density over space and find a change in density between areas of high density and areas of low density. The filled isopleths always offered the highest amounts of support for these tasks. Following Bertin's theory of visual variables, colour value and lightness are uniquely effective for: communicating ordered quantitative information because they enable users to almost intuitively sense the difference between volumes (as reviewed in Carpendale, 2003). Filled Isopleths take advantage of these qualities to visualize a field of population density in a continuous visual field, allowing the user to identify spatial changes without having to think extensively. This is also shown by Cebrykow (2025), who found that continuous visual representations of interpolated surfaces lead to better user interpretation performance than discretized surfaces do. Dot Density maps were harder to interpret, and high and low areas could sometimes be identified, but transitions between high and low areas of density often needed to be estimated by the participants. These estimates add to the uncertainty and to the cognitive workload, especially in the case of low gradients. It was more difficult to fill in the blanks in interpretations. Contour lines are used to indicate density levels explicitly; the user has to interpolate the surface between contour lines. This extra level of interpretation may account for the lesser accuracy and increased perceived difficulty of the gradient-related tasks. The results are no surprise as the basic idea of gradient interpretation is to grasp the nature of the changes being made rather than cluster defining. Representations that communicate continuous variation thus offer more support for this type of task.

4.4 Real-World Versus Synthetic Datasets

The thematic map testing was carried out on both synthetic and real-world population datasets, offering the opportunity to determine thematic map performance at varying levels of spatial complexity. In general, the participants did better on synthetic datasets than on real world datasets. This is a typical problem when dealing with cartographic perception research. Real world datasets like the population of the Ruhr region are difficult to interpret because of noise, overlapping "urban centres", and irregular spatial structures, while "simplified" datasets are often carefully designed to highlight certain spatial structures. The user studies have demonstrated that the complexity of the visual and informative elements of the map can negatively affect the map-reading performance of the map user, and can increase the cognitive load. The increased complexity of visual and informational elements of the map has been repeatedly shown to decrease map-reading performance and increase the cognitive load of the map user (Korycka – Skorupa & Gołębiowska, 2020).

The performance ranking of the three map types, however, was pretty consistent between both the data sets. The Filled Isopleths generally performed the best and were the least time consuming of the maps to respond, and Dot Density maps were also time sensitive. The consistency of the results with both the dataset type, and across the different representations, bolsters confidence that differences across map representations is related to the map representation and not to the specific dataset. These results further support the usefulness of the study as they show that the observed benefits of Filled Isopleths hold true beyond the simplified experimental scenarios and hold true in the case of realistic population distributions.

4.5 Response Time and Cognitive Efficiency

The response time is important indicator of cognitive effort needed to interpret various map types. While accuracy is an indicator of effectiveness, response time is related to efficiency. The results showed an intriguing speed/accuracy trade-off. Occasionally, Dot Density maps yielded a more rapid response; indicating that participants could quickly detect visual patterns through point distribution. This performance difference was not necessarily reflected in accuracy, however. With Dot Density maps, participants first tended to use the quick visual identification of clusters of points. But estimating density differences for each point probably would have necessitated extra visual judgement, thus compromising on the level of interpretation. However, significantly more inspection time was needed for Filled Isopleths as participants had to process continuous colour information but this extra processing does seem to have helped their decisions to be made more reliably.

However, the filled Isopleths were more likely to be inspected for a slightly longer period of time but with a much higher accuracy. In terms of usability, a compromise could be the better side since the slightly longer interpretation time led to much better performance.

Unfilled Isopleths tended to take the longest for the response. This finding is in line with their higher difficulty ratings and less accuracy scores. There was probably more time required for the participants to decipher the contour structures and to estimate the density values between contour lines.

The finding differs from those of other studies (Słomska-Przech and Gołębiowska, 2021; Sukraini et al., 2022) which concluded that choropleth maps had the highest accuracy, the fastest response time and the least difficult type of thematical map amongst the maps tested. In the present investigation, there was no single representation that emerged as the most prominent in terms of usability.

4.6 Perceived Difficulty and Objective Performance

An additional goal of this research was to evaluate perceived difficulty of the cognitive task and its relation to objective measures of difficulty. There was a clear relationship between subjective difficulty ratings and objective task performance. Generally, the tasks were rated as more difficult had lower accuracy and longer times.

This result is significant in particular, as several previous cartographic studies showed discrepancies between subjective and objective measures. Familiar maps were sometimes chosen despite their lack in effectiveness even in the best performance outcomes: for example, users indicated that they would like to use familiar map types even if they were not the most effective in terms of performance outcome. In thematic map usability studies, similar relationships have been found between evaluations and objective measures. The map types associated with higher accuracy were also found to be more highly evaluated in terms of usability (Słomska-Przech and Gołębiowska, 2021), implying that users can identify the maps that help them perform tasks efficiently.

Filled Isopleths were typically found to be easier to perceive, more accurate and to have favourable response times. The agreement between subjective and objective measures indicates good subjective assessment of the relative ease of map interpretation. The agreement between subjective and objective measures suggests that the participants were broadly successful at providing a self-representation of the cognitive challenge of each thematic representation.

The results, therefore, validate the idea that cognitive workload is an important factor in the process of interpreting thematic maps. Representations which minimize the need to interpolate between mental images and estimate visually seem to promote improved performance and more positive user experiences.

4.7 Influence of Participants Characteristics

Exploratory analyses showed that age group, gender, education and self-reported cartographic experience did not significantly relate to overall task accuracy. A lack of demographic effects was especially striking as they are often reported in previous studies for various individual user characteristics (spatial literacy, gender, and previous experience) and their connection to map reading performance (Havelková & Hanus, 2019). This study deviates from others, one reason for this might be the experimental setup. Contestants were given comprehensive written instructions and explanation with examples and practice material on how each map type should be interpreted before the survey was undertaken.

The use of these introductory materials may have minimized the differences in performance among participants because they set a foundation of shared understanding of the basics of cartography, enabling the less experienced participants to do as well as the more experienced ones. This is in keeping with Pan and Carpenter (2023) who drew on a review of cognitive frameworks to identify how initial exposures, pre-tests, and introductory prompts can be effective priming mechanisms for reducing baseline variances and successfully minimizing variances. among learners. Thus, the findings indicate that the differences found in performance were most likely due to the map depictions used, and not to the participants' characteristics.

4.8 Implications for Cartographic Design

This study's results are applicable in several ways to the design of thematic maps. The first is that Filled Isopleths are very suitable for communicating continuous patterns of population density and for identifying hotspots and gradients. They're accurate and not very difficult to perform so they're a good option for a general audience.

Second, Dot Density maps are useful when the goal is to communicate local clustering, and to give an intuitive representation of population distribution. While they were not as accurate as Filled Isopleths, fast interpretation times indicated that they were still useful for exploratory visualization.

Finally, when accurate interpretation of population density is necessary, Unfilled Isopleths may seem less appropriate for non-expert readers. Several tasks showed a drop in performance when compared to the other tasks because of the increased cognitive load of contour.

The findings of this study are consistent with the findings of many other studies in the field of cartographic usability research that have been conducted in recent years, all pointing to the need to tailor the design of thematic maps to the purpose of the analysis. The results of the present study, as well as those of previous user studies (Korycka–Skorupa & Gołębiowska, 2020; Słomska-Przech & Gołębiowska, 2021; Shaito et al., 2022) indicate that the selection of a thematic map significantly affects spatial perception and interpretation. The results thus support the notion that aesthetic considerations are not the only ones that should inform cartographic design decisions, but also the analytical tasks expected by users.

The results indicate that it is important to choose the right thematic representation based on the analytical task and that there is no single map type that is best for all analytical tasks. This re-emphasises the value of evidence-based cartographic design: the selection of maps is based on empirical assessment of user and not convention alone.

4.7 Study Limitation

This study offers important insight into the effect of various thematic map representations on the interpretation of the pattern of population density, but there are a few limitations to note.

Firstly, the experiment was carried out online using a survey, so participants did the tasks with their own devices and in different settings. So, it can be concluded that factors such as dimensions of the screen, quality of the screen, lighting and distraction were beyond the control of the experimenter and may have affected each performer's performance.

Secondly, this study was limited to analyzing the three thematic mapping techniques which were used to represent the population density such as Dot Density maps, Filled Isopleths, and Unfilled Isopleths. Thus, the results should not be extrapolated to other thematic map types and to other geographic variables without additional research.

Lastly, while the use of both synthetic and real-world-datasets added value to the study, through controlled and realistic assessment, the synthetic datasets were purposefully designed to emphasize specific hotspot and gradient patterns. As a result, they are unable to adequately portray the richness of geographic features in the real world.

Overall, the results have been obtained from a controlled experimental design, multiple usability measures, synthetic and real-world datasets, indicating confidence in the results contributing to the cartographic perception research and practical guidance for the design of thematic maps.

5.CONCLUSION

The user study conducted revealed the different interpretations of spatial patterns of population density based on the type of maps presented. Generally, the filled Isopleths yielded the most accurate and least-difficult results, and Dot Density maps allowed for quicker responses. The unfilled Isopleths correlated with greater difficulty and lower performance for a number of tasks.

The findings also showed that the effectiveness of maps is dependent upon the task that is being done. Filled Isopleth performed well for creating continuous density patterns and aiding with gradient interpretation while Dot Density maps continued to be useful for identifying local elevations and clustering patterns. Both synthetic and real-world datasets demonstrated that these patterns held true in a variety of spatial settings.

The collected data was generally in favor of Filled Isopleth, but the results do not imply that in all cases such a type of map should be used. There are certain advantages of each representation depending on the map and the information conveyed. Thus, the choosing of the thematic mapping technique should be directed by the analytical job, the properties of the data and the requirements of the target audience.

References

[1] Ashour, L. (2024). A review of user-friendly freely-available statistical analysis software for medical researchers and biostatisticians. *Research in Statistics*, *2*(1). https://doi.org/10.1080/27684520.2024.2322630

[2] Beitlova, M., Popelka, S., & Vozenilek, V. (2020). Differences in Thematic Map Reading by Students and Their Geography Teacher. *ISPRS International Journal of Geo-Information*, *9*(9), 492. https://doi.org/10.3390/ijgi9090492

[3] Bertin, J. (1983). *Semiology of graphics; diagrams networks maps.* (University of Wisconsin, Press, Madison,

[4] Besançon, L., Cooper, M., Ynnerman, A., & Vernier, F. (2020). *An evaluation of visualization methods for population statistics based on choropleths maps*. arXiv. https://arxiv.org/abs/2005.00324

[5] Bondarenko M., Priyatikanto R., Tejedor-Garavito N., Zhang W., McKeen T., Cunnigham A., Woods T., Hilton J., Cihan D., Nosatiuk B., Brinkhoff T., Tatem A., Sorichetta A., *Constrained estimates of 2015-2030 total number of people per grid square at a resolution of 3 arc (approximately 100m at the equator) R2025A version v1. Global Demographic Data project – Funded by The Bill and Melinda Gates Foundation (INV-045237). WorldPop – School of Geography and Environmental Science, University of Southampton*. Doi: 10:5258/SOTON/WP00839

[6] Carpendale, M.S.T. (2003). *Considering Visual Variables as a Basis for Information Visualisation* (Technical Report 2003-734-37). Department of Computer Science, University of Calgary.

[7] Cebrykow, M. (2025). Using and isopleth map to determine the population of any area. *Journal of Geography, Politics and Society. 15(1),* 106-115

[8] Çöltekin, A., Heil, B., Garlandini, S., & Fabrikant, S. I. (2009). Evaluating the Effectiveness of Interactive Map Interface Designs: A Case Study Integrating Usability Metrics with Eye-Movement Analysis. *Cartography and Geographic Information Science*, *36*(1), 5–17. https://doi.org/10.1559/152304009787340197

[9] Crump MJC, McDonnell JV, Gureckis TM (2013) Evaluating Amazon's Mechanical Turk as a Tool for Experimental Behavioural Research. PLOS ONE 8(3): e57410. https://doi.org/10.1371/journal.pone.0057410

[10] Ding, L., & Meng, L. (2014). A comparative study of thematic mapping and scientific visualization. *Annals of GIS*, *20*(1), 23–37. https://doi.org/10.1080/19475683.2013.862856

[11] Ehrlich, D., Kemper, T., Pesaresi, M., & Corbane, C. (2018). Built-up area and population density: Two essential societal variables to address climate hazard impact. *Environmental Science & Policy, 90, 73-82.* https://doi.org/10.1016/j.envsci.2018.10.001

[12] Forest, D., & Medyńska-Gulij, B (2021). *Which mapping technique for population density is effective, attractive, and suggestive?* Abstracts of the International Cartographic Association, 3, 1. https://doi.org/10.5194/ica-abs-3-82-2021

[13] GDAM. (2024). Database of Global Administrative Areas (Version 4.1). Retrieved October 12, 2025, from https://gadm.org

[14] Havelková, L., & Hanus, M. (2019). Map Skills in Education: A Systematic Review of Terminology, Methodology, and Influencing Factors. *Review of International Geographical Education Online*, *9*(2), 361-401. https://doi.org/10.33403/rigeo.591094

[15] Korycka-Skorupa, J., & Gołębiowska, I. M. (2020). Numbers on Thematic Maps: Helpful Simplicity or Too Raw to Be Useful for Map Reading? *ISPRS International Journal of Geo-Information*, *9*(7), 415. https://doi.org/10.3390/ijgi9070415

[16] Lazar, J., Feng, J. H., & Hochheiser, H. (2017). Research methods in human-computer interaction (2nd ed.). Morgan Kaufmann Publishers.

[17] M. He, X. Tang, and Y. Huang, "To visualize spatial data using thematic maps combined with infographics," *2011 19th International Conference on Geoinformatics*, Shanghai, China, 2011, pp. 1-5, Doi: 10.1109/GeoInformatics.2011.5980880.

[18] MacEachren, A.M. (2004). *How maps work: Representation, Visualization, and design (2nd ed.).* Guilford Press

[19] Medyńsk a-Gulij, B., Wielebski, Ł., Halik, Ł., & Smaczyński, M. (2020). Complexity Level of People Gathering Presentation on an Animated Map—Objective Effectiveness Versus Expert Opinion. *ISPRS International Journal of Geo-Information*, *9*(2), 117. https://doi.org/10.3390/ijgi9020117

[20] Pan, S.C., Carpenter, S.K. Prequestioning and Pretesting Effects: A Review of Empirical Research, Theoretical Perspectives, and Implications for Educational

Practice. *Educ Psychol Rev* **35**, 97 (2023). https://doi.org/10.1007/s10648-023-09814-5

[21] Peer, E., Brandimarte, L., Samat, S., & Acquisti, A. (2017). Beyond the Turk: Alternative platforms for crowdsourcing behavioural research. *Journal of Experimental Social Psychology, 70, 153–163.* https://doi.org/10.1016/j.jesp.2017.01.006

[22] Peters, S. and Meng, L., 2015. Spatial-temporal density mapping for spatially extended dynamic phenomena – a novel approach to incorporate movements in density maps. *International Journal on Advances in Intelligent Systems, 8(1 & 2), pp.27-39.*

[23] Pickle, L. W. (2003). Usability testing of map designs. In Proceedings of the 22nd International Cartographic Conference (ICC 2003). International Cartographic Association.

[24] R core Team (2025). R: *A Language and environment for statistical computing.* (Version 4.5) [Computer software]. Retrieved from https://cran.r-project.org. (R packages retrieved from CRAN snapshot 2025-05-25)

[25] Roth, R. E. (2013). Interactive maps: What we know and what we need to know. *Journal of Spatial Information Science, (6), 59–115.* https://doi.org/10.5311/JOSIS.2013.6.105

[26] Roth, R. E., Çöltekin, A., Delazari, L., Filho, H. F., Griffin, A., Hall, A., … van Elzakker, C. P. J. M. (2017). User studies in cartography: opportunities for empirical research on interactive maps and visualizations. *International Journal of Cartography*, *3*(sup1), 61–89. https://doi.org/10.1080/23729333.2017.1288534

[27] Roth, R. E., Kelly, M., Underwood, N., Vincent, K., & Sack, C. (2019). Interactive & Multiscale Thematic Maps: A preliminary Study. *Abstracts of the ICA,* 1, 315.https://doi.org/10.5194/ica-abs-1-315-2019

[28] Roth, R.E., Kelly, M., Underwood, N., Lally, N., Liu, X., Vincent, K., & Sack, C. (2020). Interactive & Multiscale Thematic Maps: Preliminary Results from an Empirical Study. Abstracts of the ICA, 2, 135. https://doi.org/10.5194/ica-abs-2-135-2020

[29] Şahin, M., & Aybek, E. (2020). Jamovi: An Easy-to-Use Statistical Software for the Social Scientists. *International Journal of Assessment Tools in Education, 6(4), 670-692*. https://doi.org/10.21449/ijate.661803

[30] Shaito M, Elmasri R et al., Comparison of Map Visualization Techniques Used for Spatial and Spatio-Temporal Data: Analytical Survey Applied to Covid-19 Data, *Medical Research Archives, [online] 10(9)* https://doi.org/10.18103/mra.v10i9.3072

[31] Silva de Souza, R., Sequeira, C. A., & Borges, E. M. (2024). Enhancing statistical education in chemistry and STEAM using JAMOVI. Part 1: Descriptive statistics and comparing independent groups. *Journal of Chemical Education, 101(12), 5027–5039.*

[32] Slocum, T.A., McMaster, R.B., Kessler, F. C., & Howard, H.H. (2023). *Thematic cartography and geovisualization* (5th ed.) CRC Press. https://doi.org/10.1201/9781003150527

[33] Słomska-Przech, K., & Gołębiowska, I. M. (2021). Do Different Map Types Support Map Reading Equally? Comparing Choropleth, Graduated Symbols, and Isoline Maps for Map Use Tasks. *ISPRS International Journal of Geo-Information, 10(2), 69.* https:/doi.org/10.3390/ijgi10020069

[34] Sukraini, T. T., Yasa, I. M. A., & Wiguna, P. P. K. (2022). Comparing choropleth and graduated symbols: How different map types affect public understanding in COVID-19 map reading in Badung Regency, Bali, Indonesia. *Geographia Technica, 17(1), 150–166.* https://doi.org/10.21163/GT_2022.171.12

[35] The jamovi project (2025). Jamovi. (Version 2.7) [Computer Software]. Retrieved from https://www.jamovi.org

[36] Traun, C., Klug, H., & Burkhard, B. (2017). Mapping techniques. In B. Burkhard & J. Maes (Eds.), *Mapping ecosystem services* (pp. 55-62). Pensoft Publishers

[37] Traun, C., Schreyer, M. L., & Wallentin, G. (2021). Empirical Insights from a Study on Outlier Preserving Value Generalization in Animated Choropleth Maps. *ISPRS International Journal of Geo-Information*, *10*(4), 208. https://doi.org/10.3390/ijgi10040208

[38] Wielebski, Ł., & Medyńska-Gulij, B. (2023). User Evaluation of Thematic Maps on Operational Areas of Rescue Helicopters. *ISPRS International Journal of Geo-Information*, *12*(2), 30. https://doi.org/10.3390/ijgi12020030

[39] Zanin, C., Lambert, N., Davoine, P. -A., Mathan, H., & Saint-Marc, C. (2013). Mapping guide. ESPON – Inspire Policy Making with Territorial Evidence. https://hal.science/hal-03594611

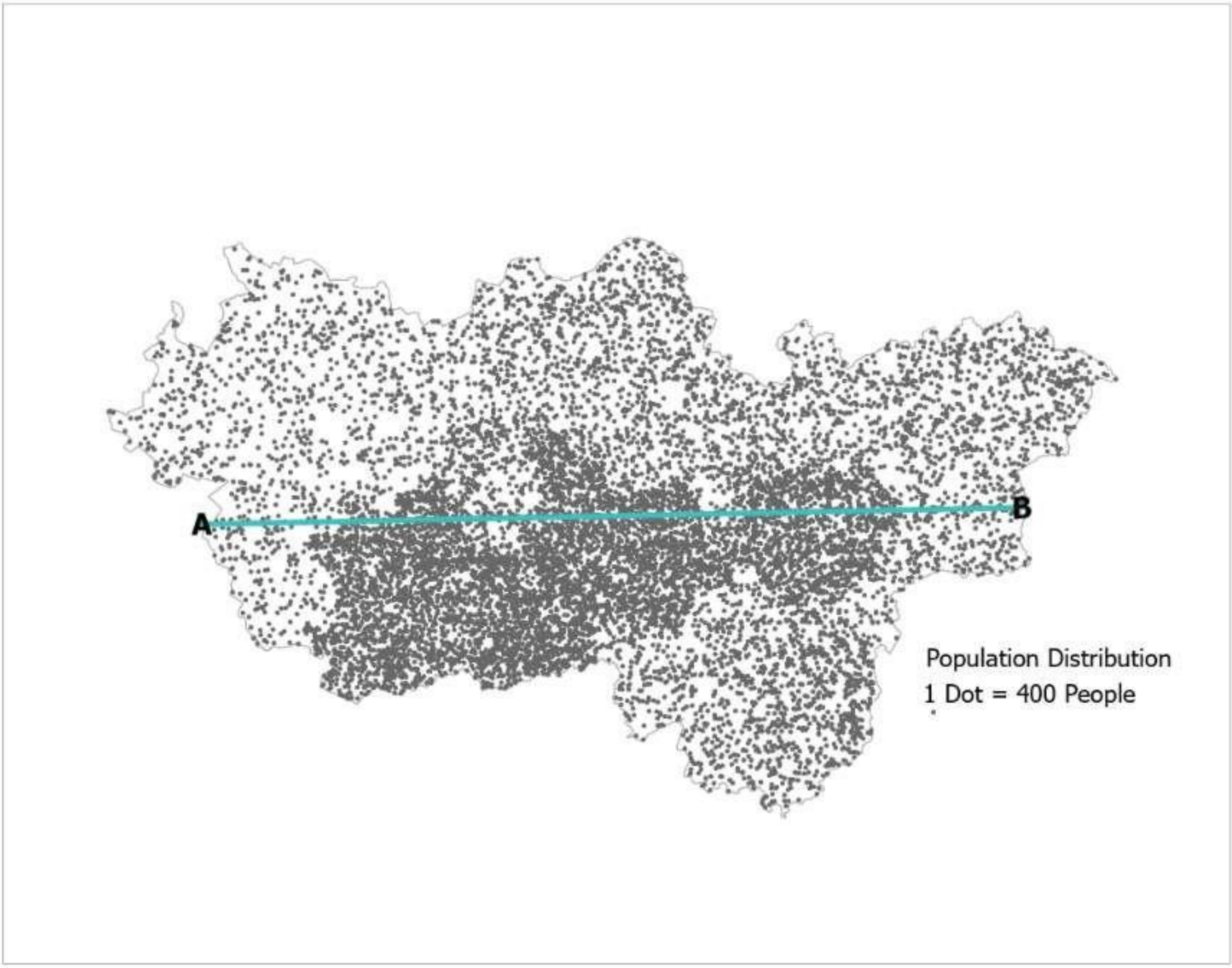
Git Repository

GitHub - sunday2023539/ASSESSMENT-OF-THEMATIC-MAPPING-TECHNIQUES-FOR-POPULATION-DISTRIBUTION-VISUALIZATION · GitHub

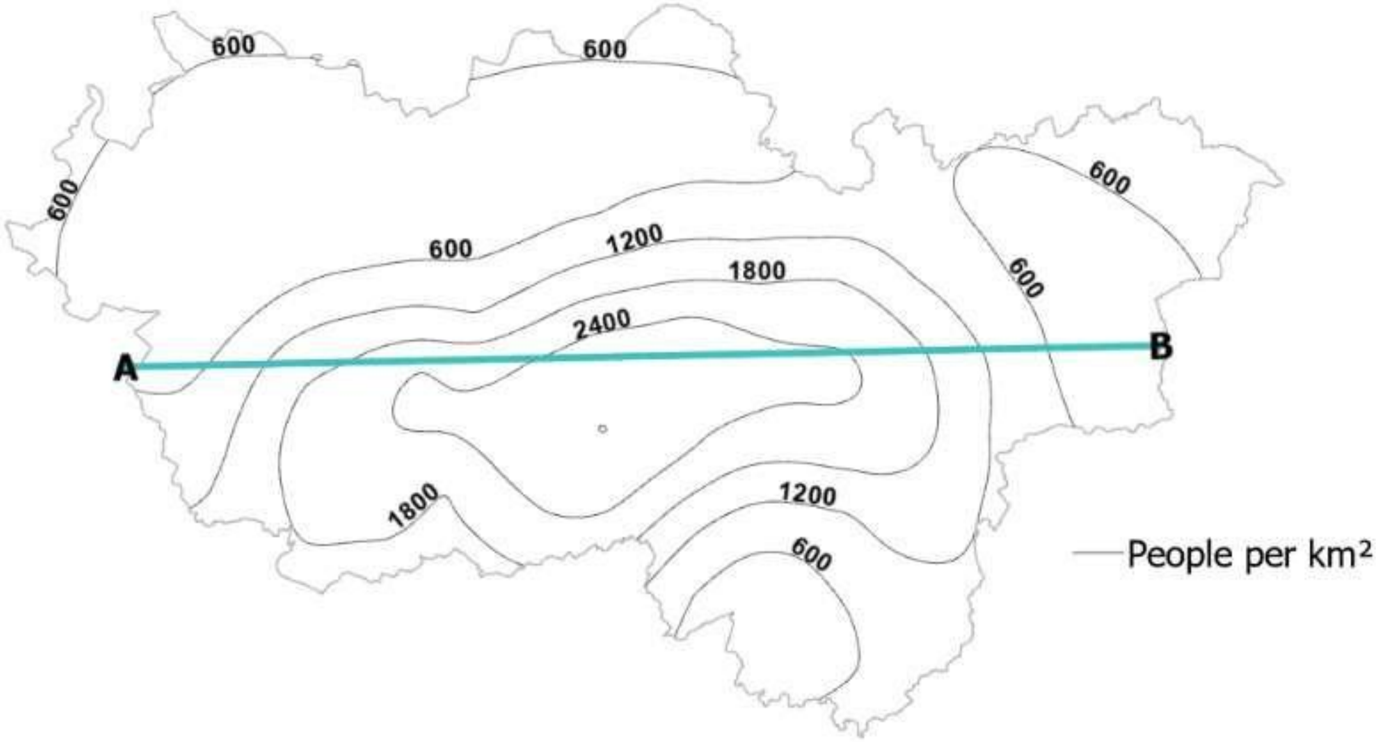
Study Survey

https://survey.plus.ac.at/index.php/624757?lang=en

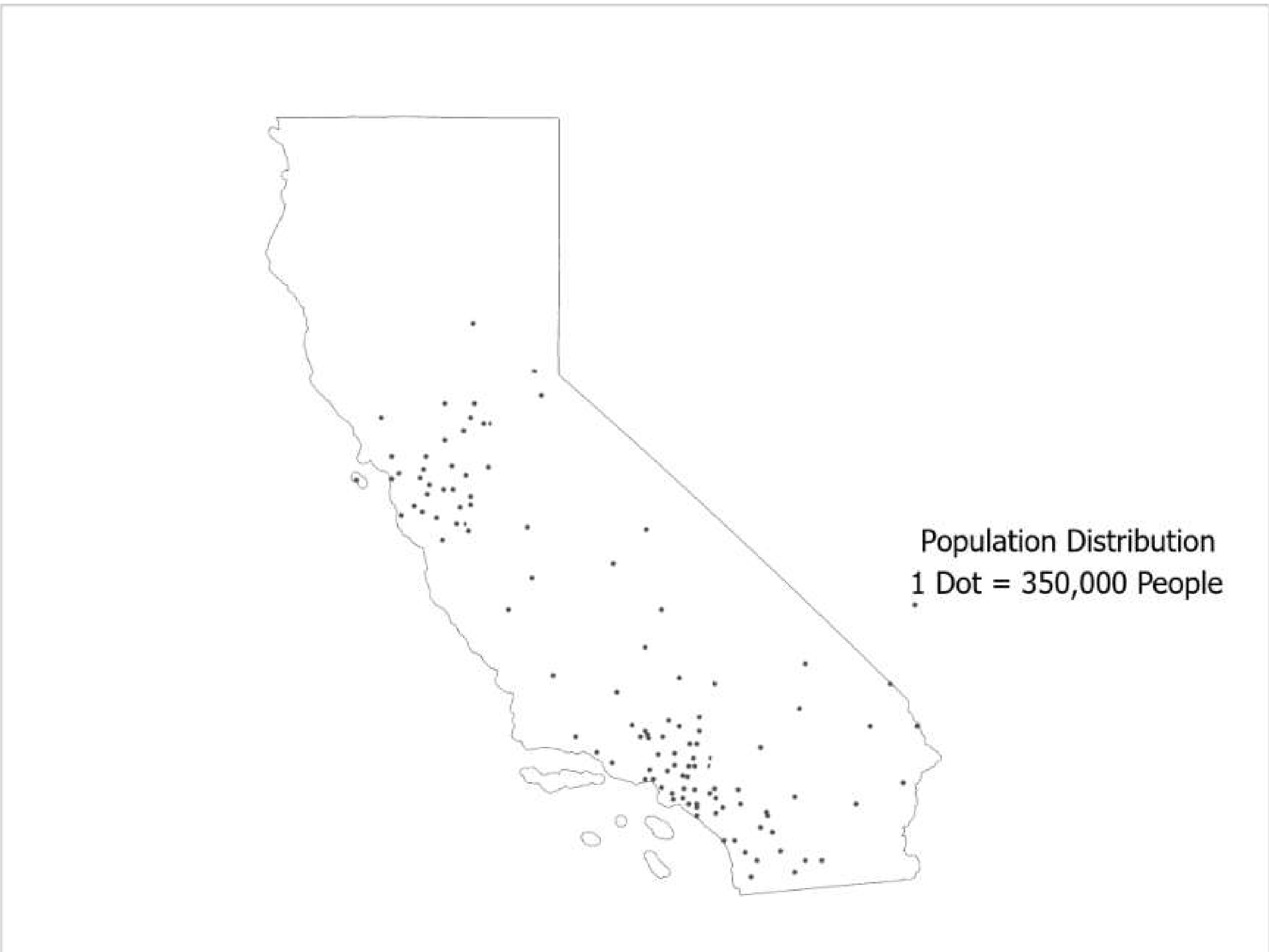
RW_GD

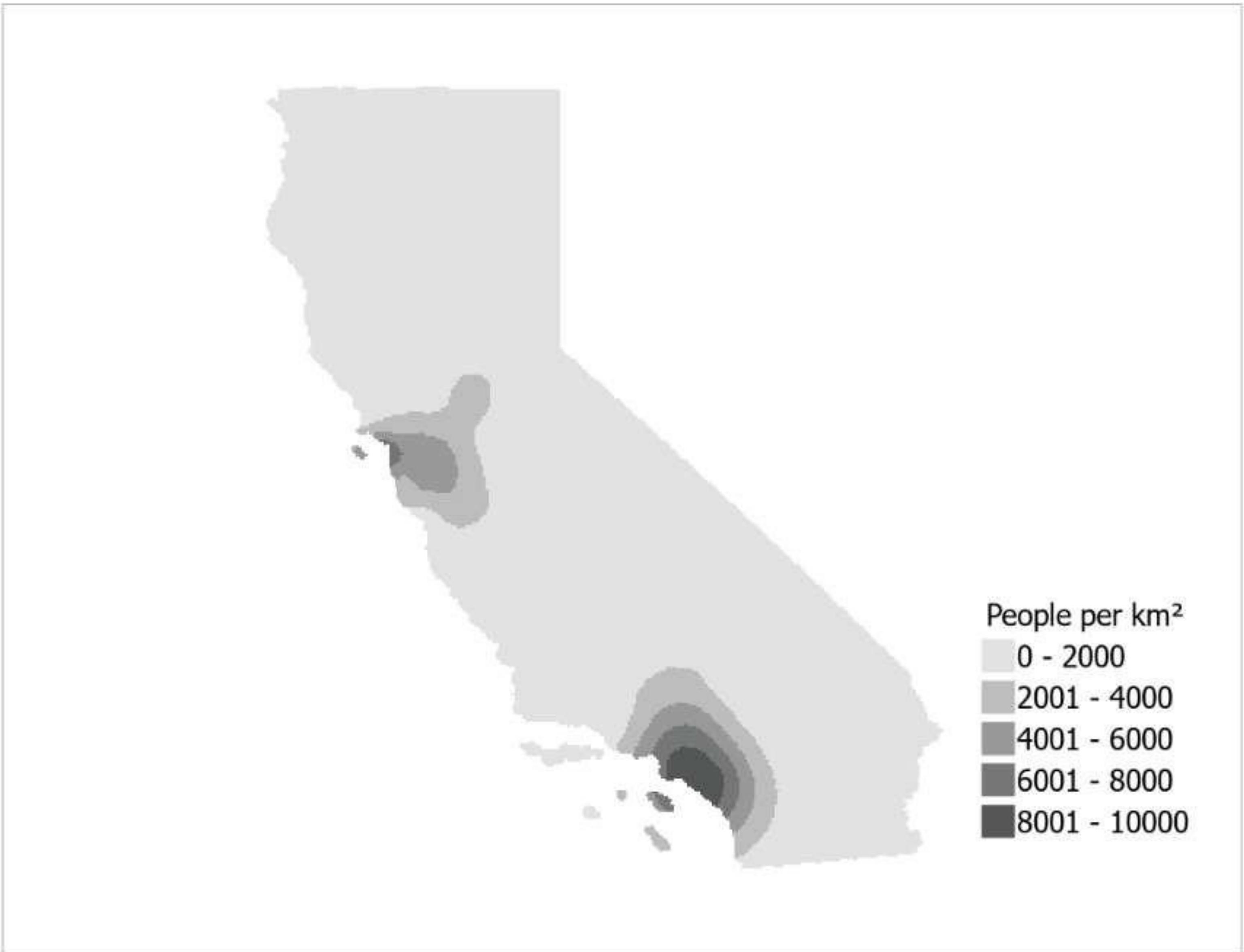



A
B
People per km²
0 - 600
601 - 1,200
1,201- 1,800
1,801 - 2,400
2,401- 3,000


600
600
600
600
1200
1800
2400
600
600
1800
1200
600
A
B
People per km²

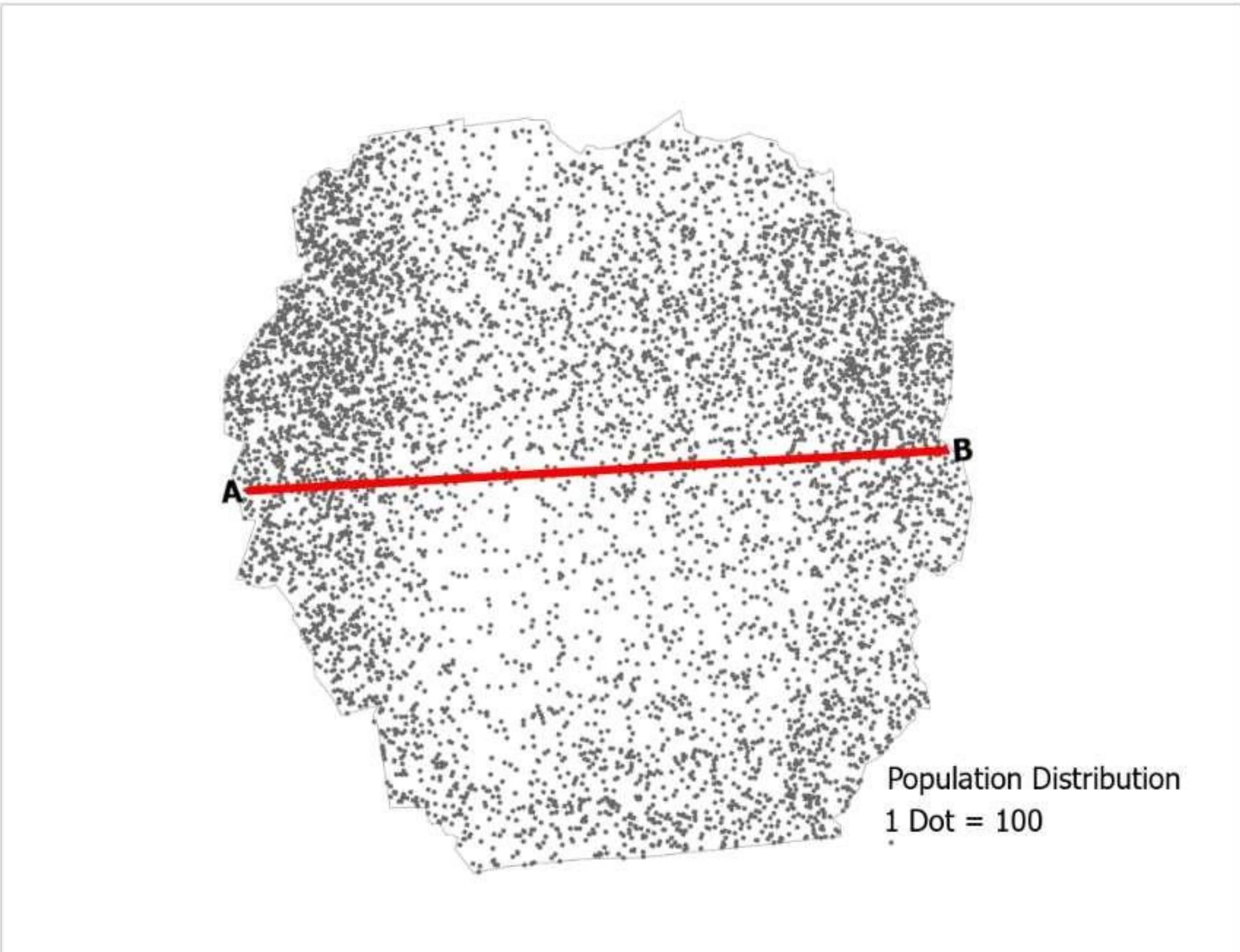
RW_HS

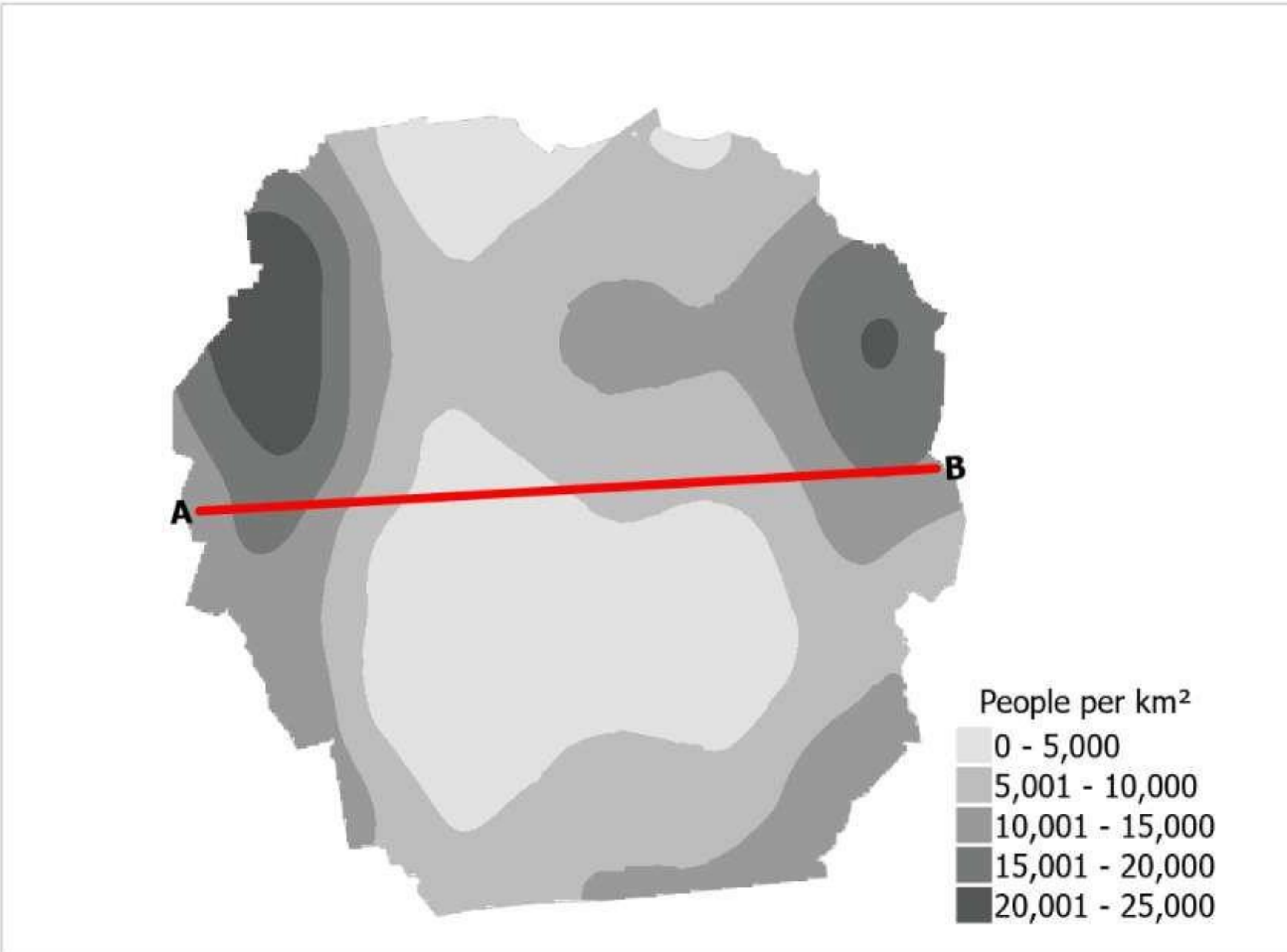


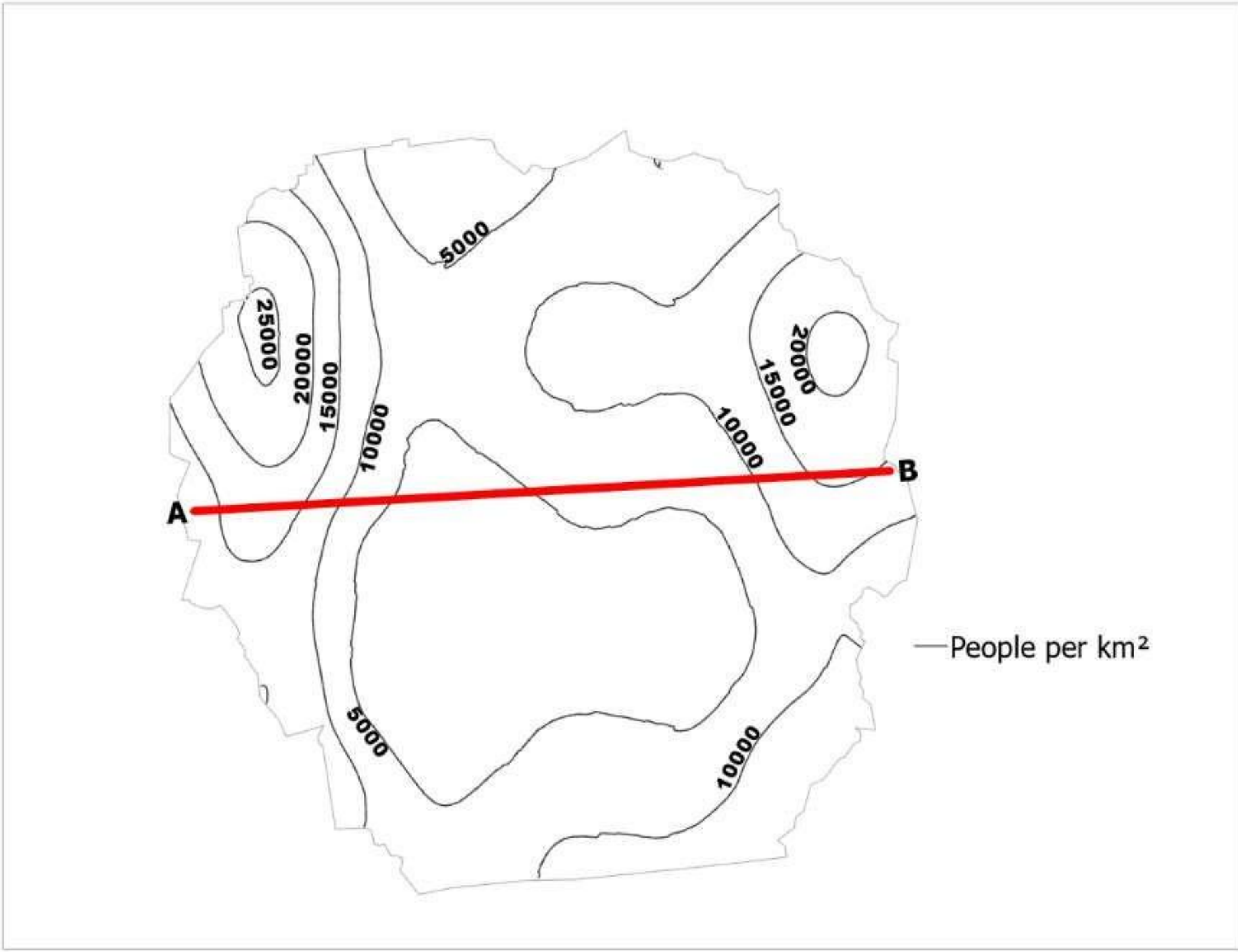

People per km²
0 - 2000
2001 - 4000
4001 - 6000
6001 - 8000
8001 - 10000


4000
2000
4000
2000
8000
6000
People per km²

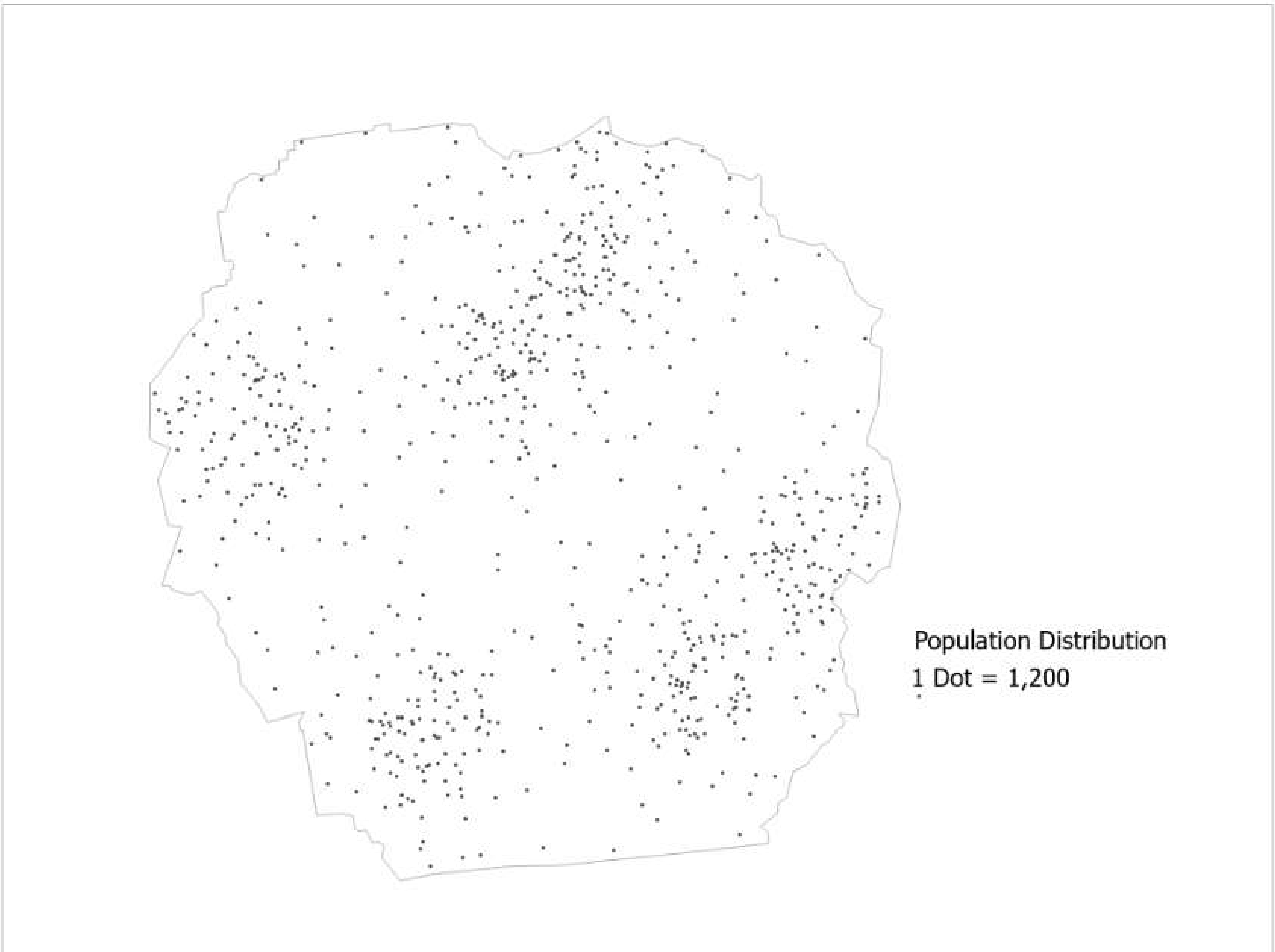
SYN_GD_A

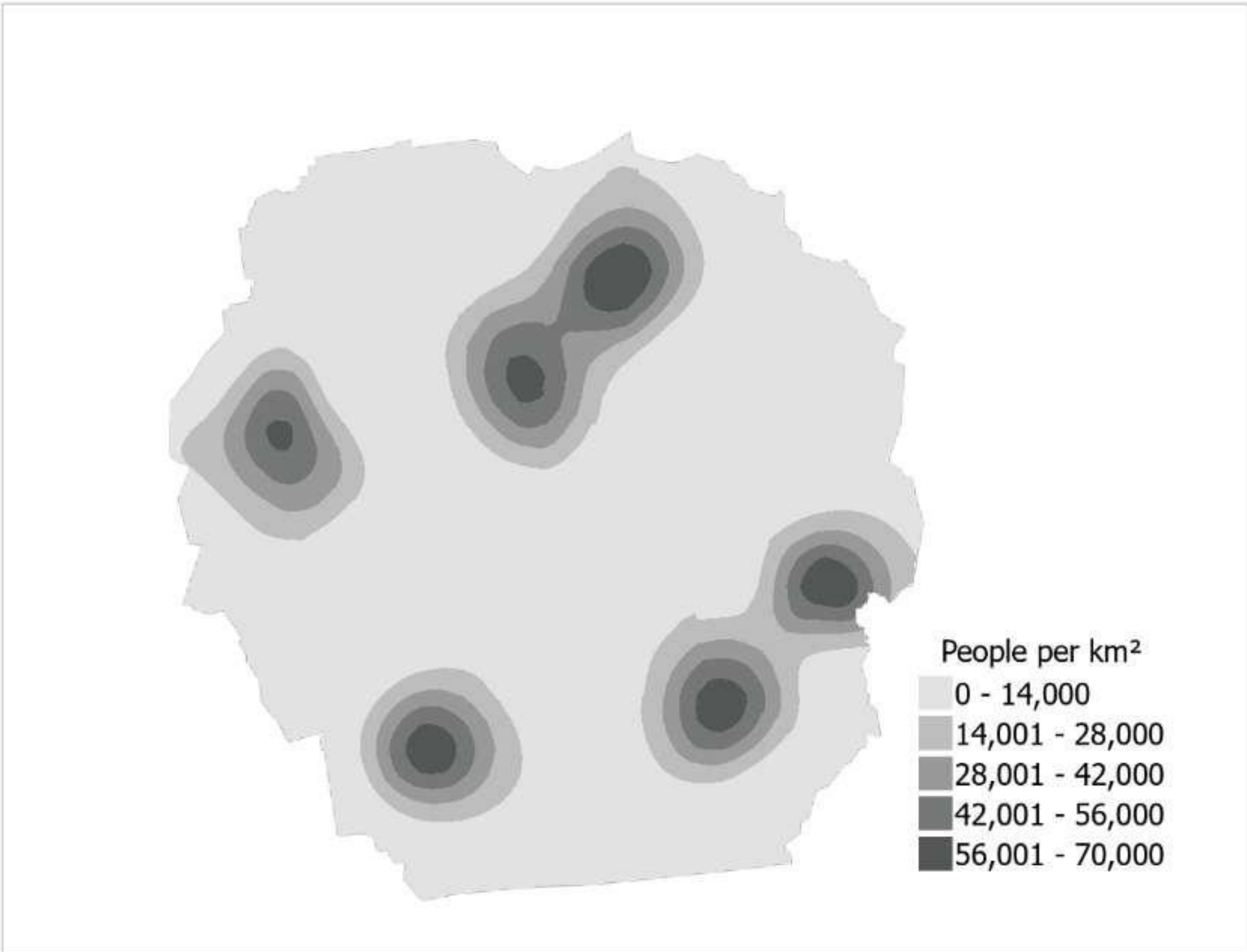


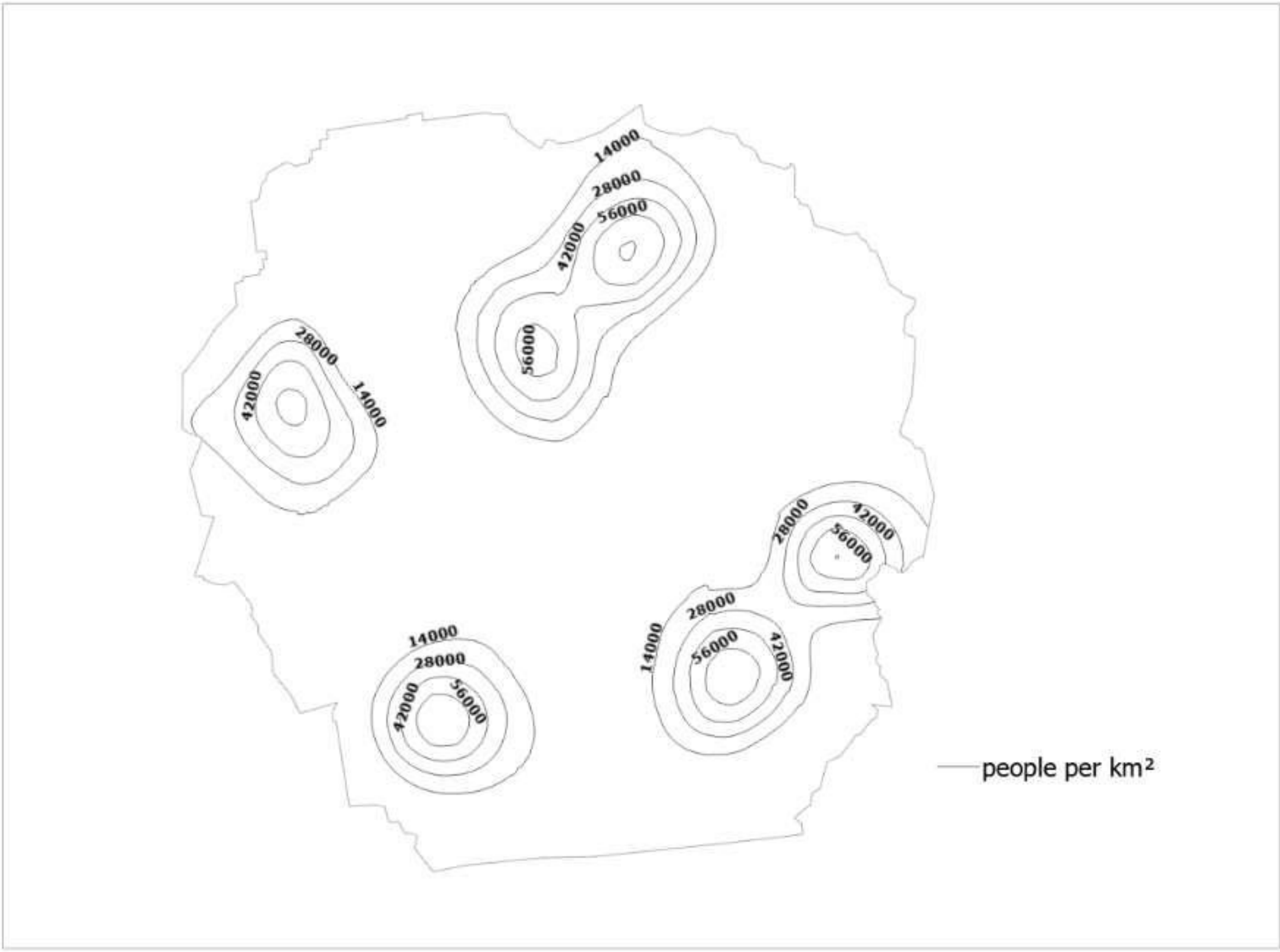

A
B
People per km²
0 - 5,000
5,001 - 10,000
10,001 - 15,000
15,001 - 20,000
20,001 - 25,000


5000
25000
20000
15000
10000
20000
15000
10000
A
B
People per km²
5000
10000

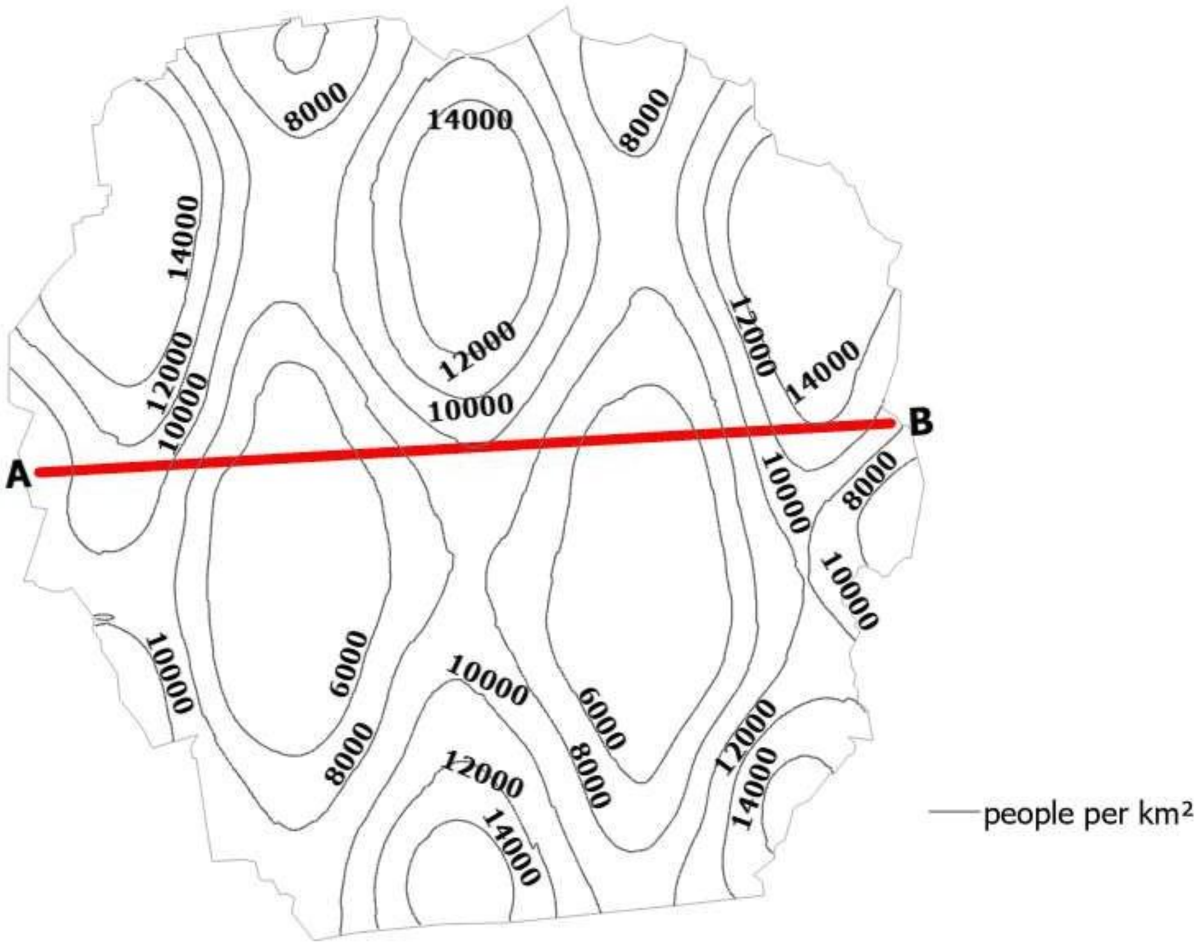
SYN_HS_A

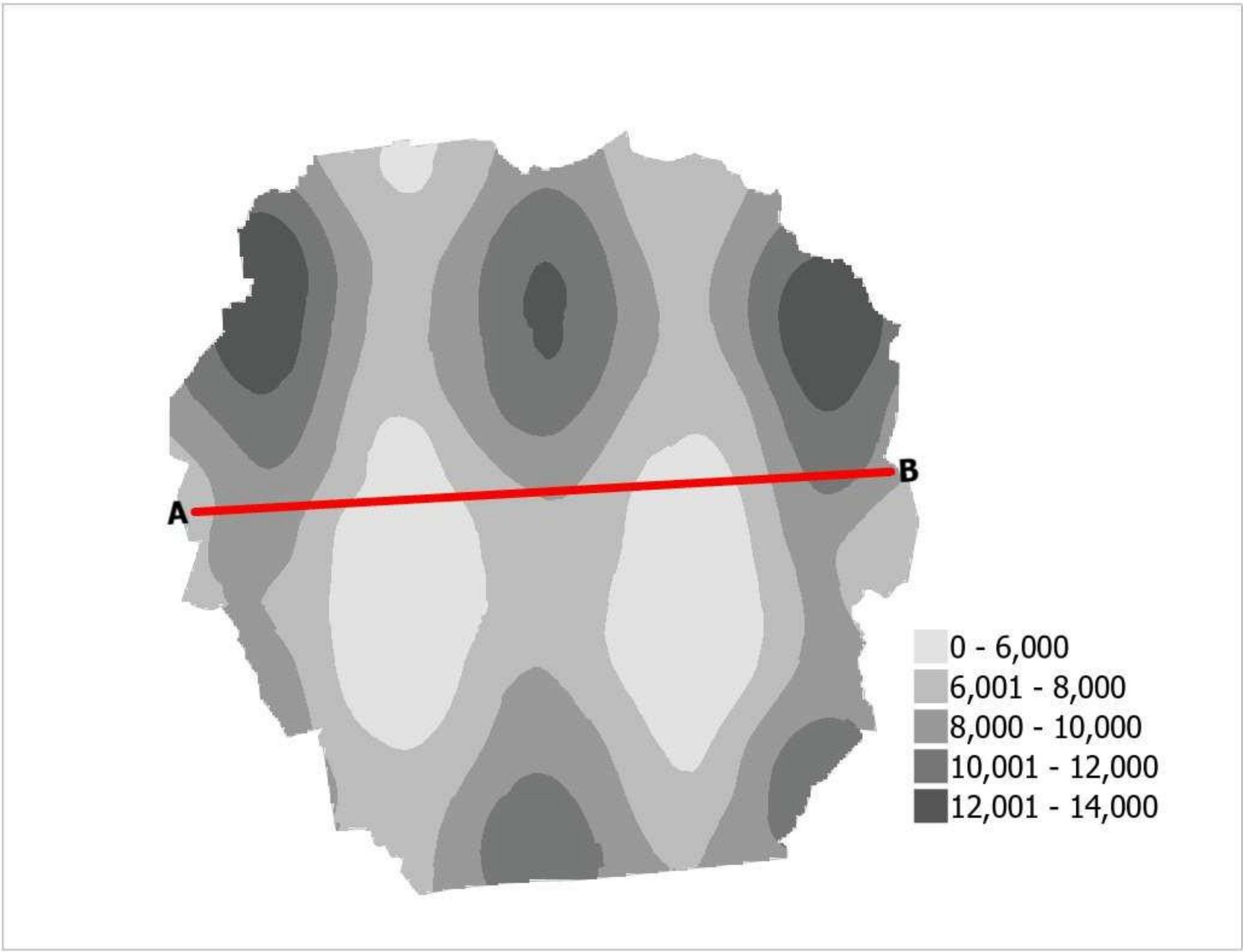


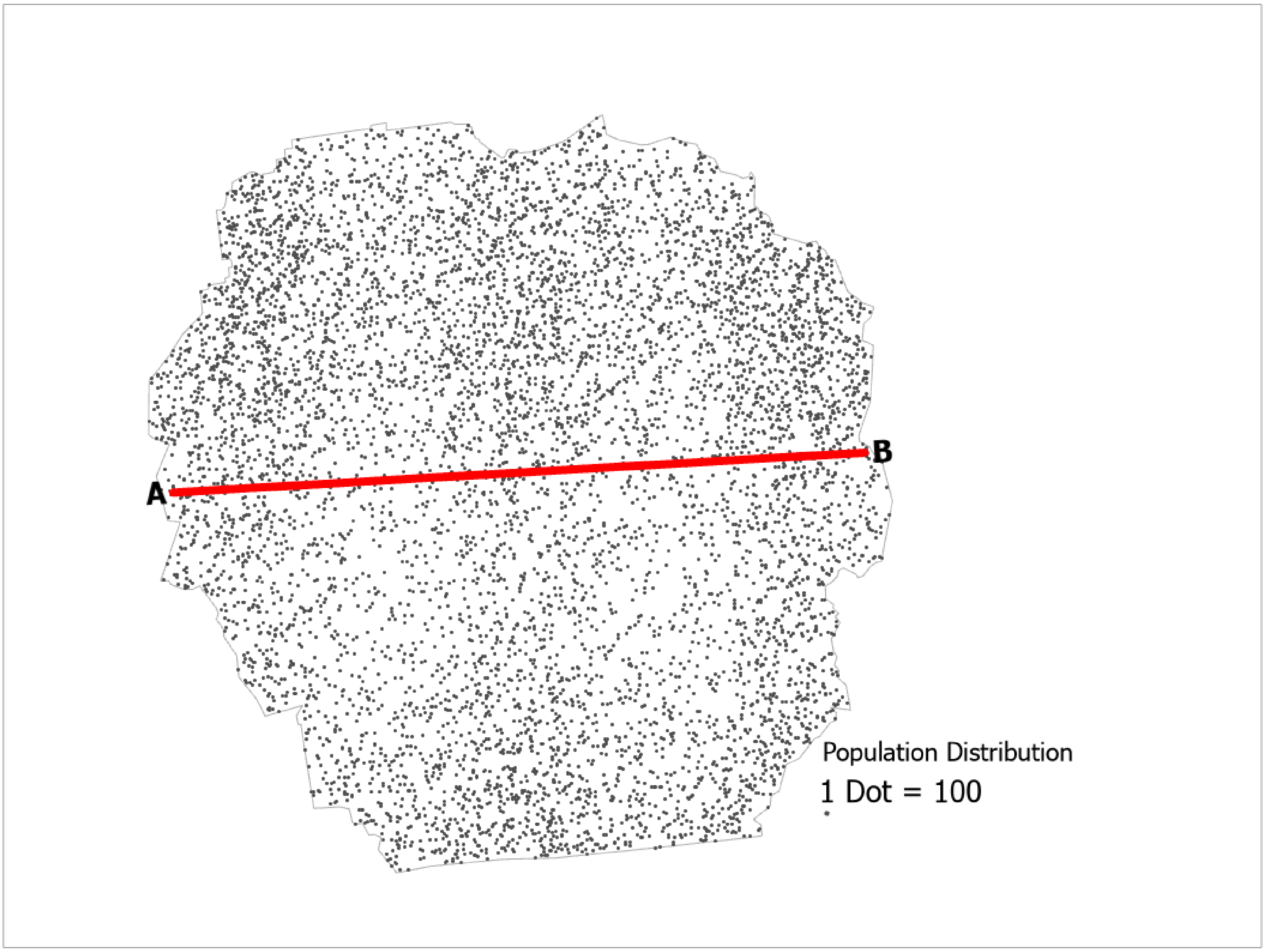

People per km²
0 - 14,000
14,001 - 28,000
28,001 - 42,000
42,001 - 56,000
56,001 - 70,000


14000
28000
56000
42000
56000
28000
42000
14000
28000
42000
56000
28000
14000
28000
56000
42000
14000
28000
42000
56000
people per km²

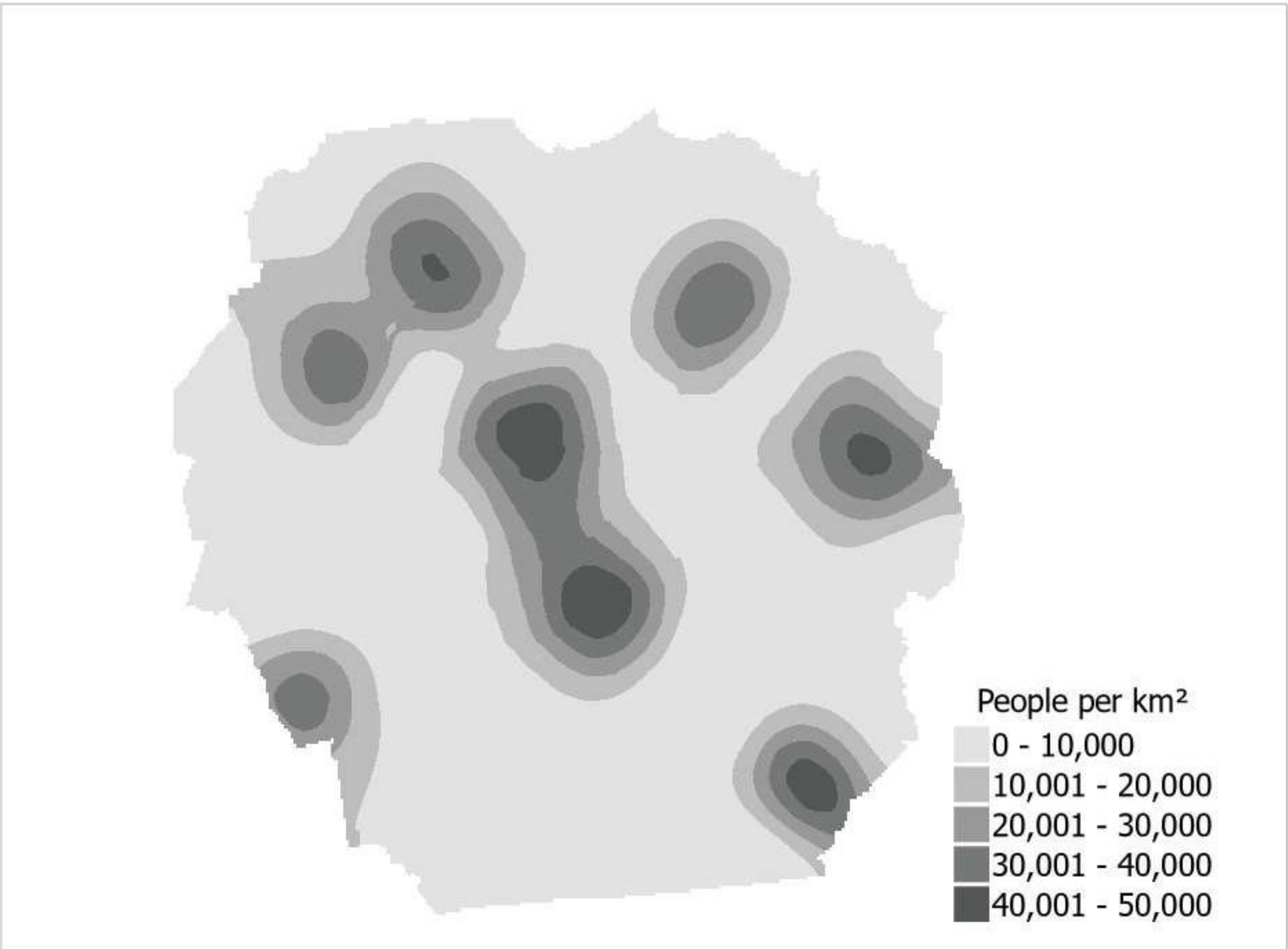
SYN_GD_B

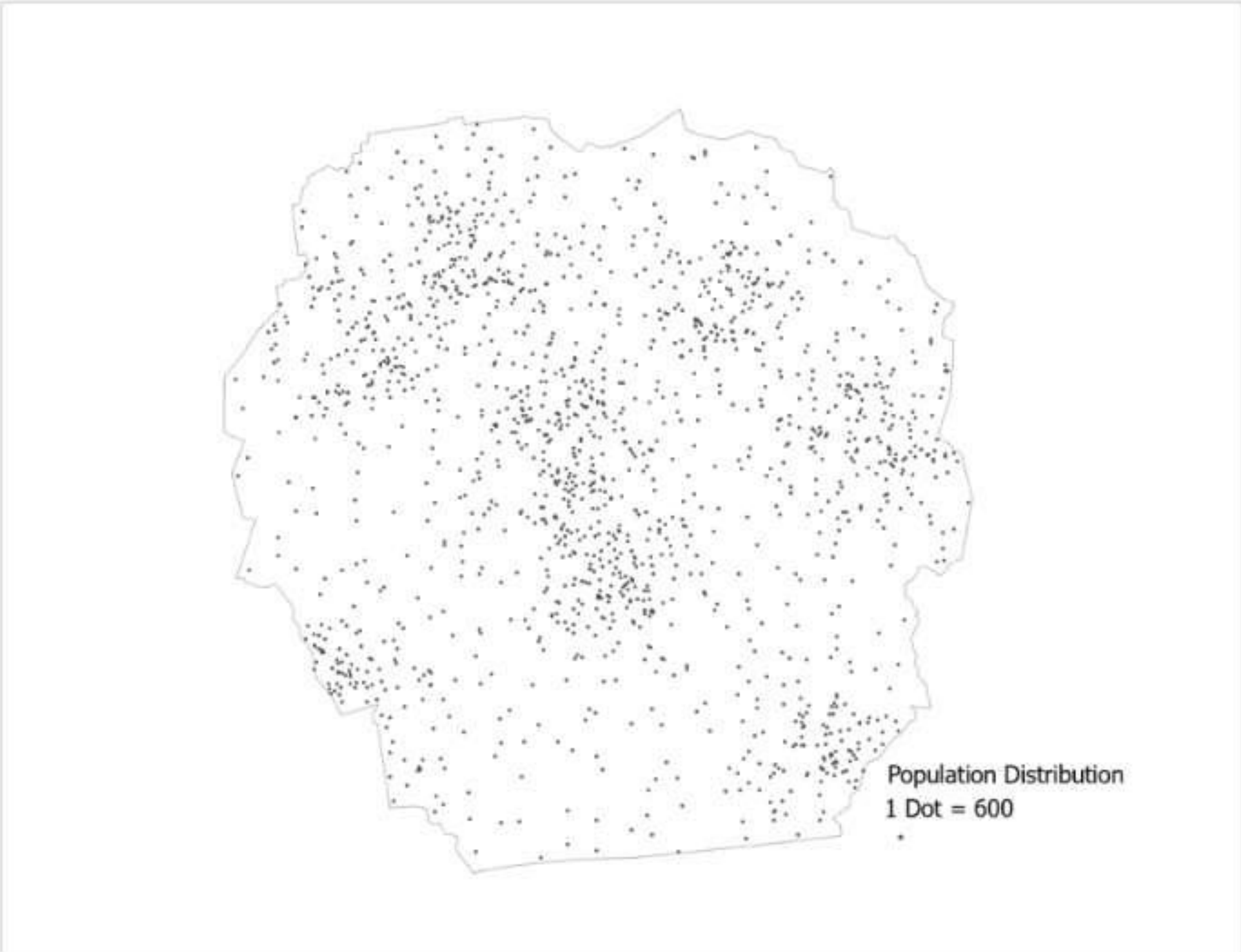

8000
14000
8000
14000
12000
10000
12000
10000
12000
14000
B
A
10000
8000
10000
10000
6000
10000
6000
8000
8000
12000
12000
14000
14000
people per km²

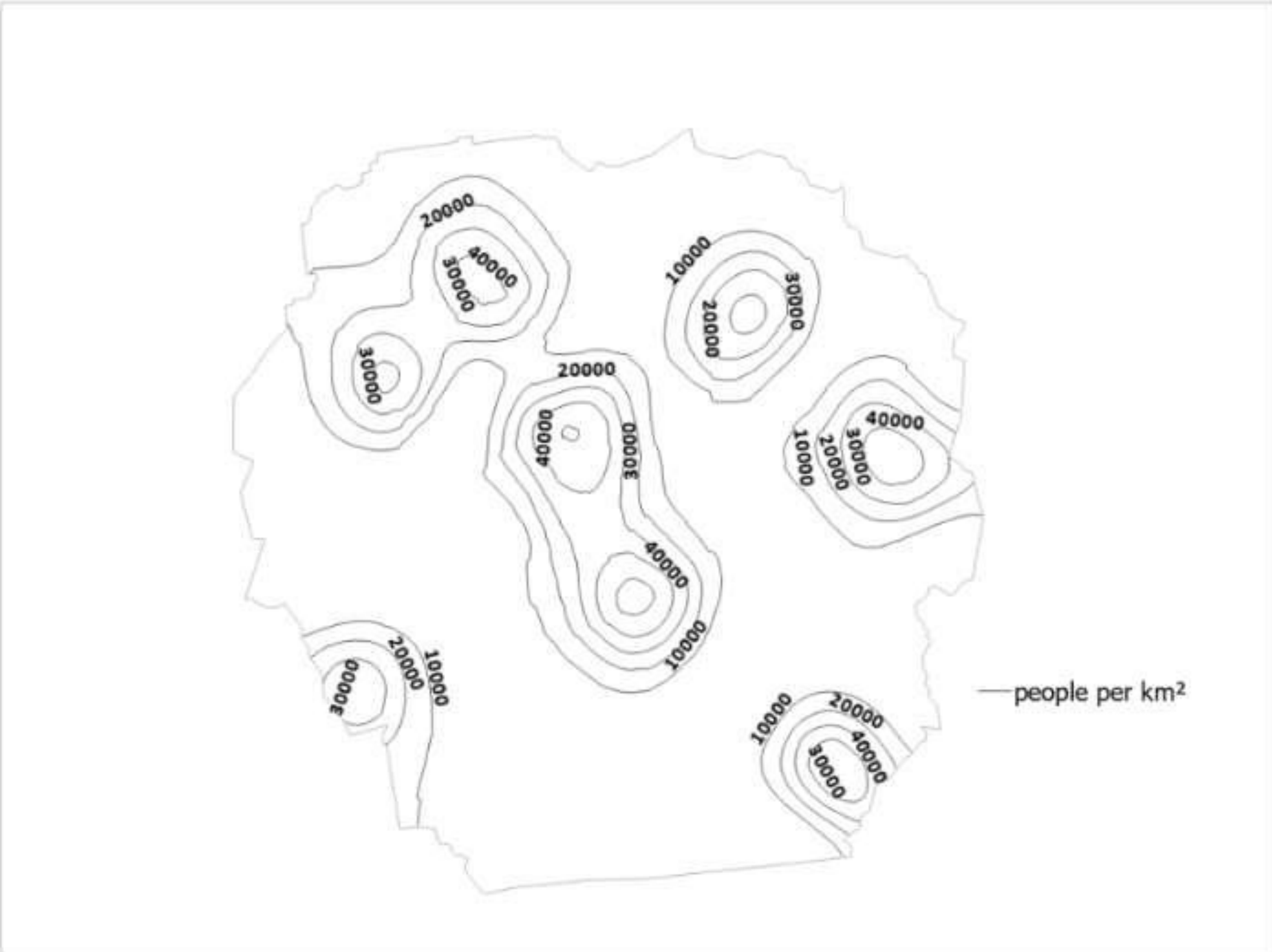

A
B
0 - 6,000
6,001 - 8,000
8,000 - 10,000
10,001 - 12,000
12,001 - 14,000


A
B
Population Distribution
1 Dot = 100

SYN_HS_B


People per km²
0 - 10,000
10,001 - 20,000
20,001 - 30,000
30,001 - 40,000
40,001 - 50,000


Population Distribution
1 Dot = 600


20000
40000
30000
30000
10000
30000
20000
20000
40000
30000
40000
10000
20000
30000
40000
10000
30000
20000
10000
10000
20000
40000
30000
—people per km²